

QUESTION FOR DISCUSSION AND RECAPITULATIONS

1. How do you prepare Lesson Plan for level 1?
2. Explain the steps involved in planning lesson plan
3. How can you prepare the four-fold lesson plan?
4. Prepare the model lesson plan the class 12th Std any commerce and accountancy topic
5. Explain Herbartian approach to lesson planning
6. What are Levels of Teaching- Memory, Understanding & Reflective Level?
7. Explain Comparison of memory level, understanding level and reflective level
8. Explain Unit Plan, Advantage of Unit Plan
9. Explain Year Plan, Advantage of Year Plan
10. What are relationship of Educational Objectives and Instructional Objective?
11. What is unit plan? how to plan as per prescribed format explain
12. What is year plan? how to plan as per prescribed format explain
13. What are lesson plan goals in teaching of commerce and accountancy
14. What are the advantages of commerce and accountancy
15. What are the steps of commerce and accountancy
16. What are Characteristics of an effective lesson plan

UNIT 4 METHODS OF TEACHING

OBJECTIVES

- ❖ Know about Commerce and Accountancy teaching
- ❖ Understand different approaches, methods and techniques in Commerce and Accountancy teaching
- ❖ Know about modern techniques in Commerce and Accountancy teaching

4.1 METHODS OF TEACHING COMMERCE AND ACCOUNTANCY

The important aim of teaching is to bring about socially desirable behaviours among the students. It can be realized through effective teaching following its basic principles. But how the students learn effectively content is depending on the method of teaching. There is a great world outside and within the mind of the child. Therefore it is the foremost duty of the teacher to bring the two together. This process of interpreting the world of knowledge to the child's mind is called method of teaching. It is first way of teaching. Method is the style of presentation of content in classrooms.

It is called that teaching is an art therefore it can be assumed that some people are born teachers. Besides these a huge number

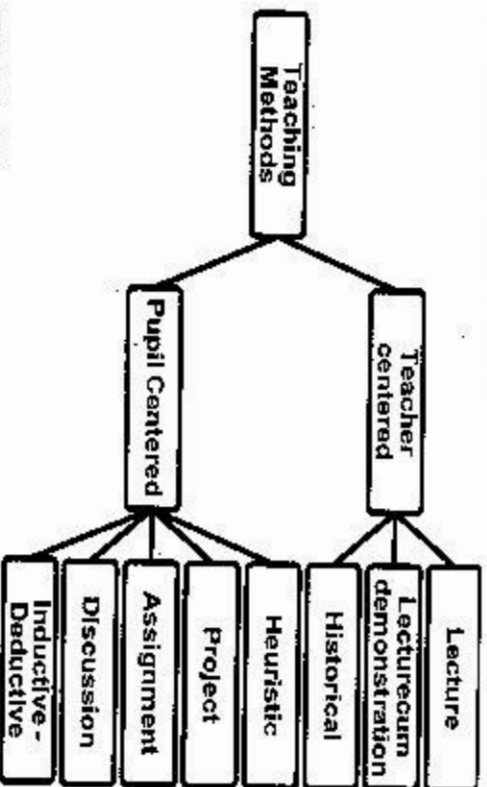
of people do not have any interest in teaching and unable to create students motivation in the subject. Still they can improve upon practice and by the following a method of teaching. Thus it can be concluded that every teacher should familiar with different methods of teaching, which are the points of discussion in the present chapter.

The method of teaching is not an end but a means to arrive the aims of teaching, therefore it is well-known fact that any one method is not feasible all the times and in all conditions while the every method do not serve the purpose of all the teachers in uniform manner.

4.1.1 Types of Teaching Methods

Teaching methods are categorized into two classes

- ❖ Teacher centered methods
- ❖ Pupil centered methods



4.2 TEACHER-CENTERED INSTRUCTION - LECTURE METHOD

In our teaching the lecture method dominates to a very great extent. The reason is that it has been far too long with us. As a matter of fact, the most teachers find lectures almost indispensable for certain purposes lecture can be used for

- ❖ Motivating large number of pupils at the same time.
- ❖ Introducing activities for groups of pupils in a class.
- ❖ For summing up at the end of the units, and

- ❖ For explaining difficult points.

They are particularly used for bridging gaps between topics to be studied in depth for the presenting information would otherwise be unavoidable to the pupils. The method is very convenient to the teacher who needs no practical preparation. In this method, the teacher occupies the central place in the teaching learning process. In spite of the money values, the formal lecture is ordinarily in ineffective method for teaching secondary school pupils. Because, the learning forms the lectures is relatively passive learning. It may be sterile learning also. Pupils learn better when they put themselves in to learning. Unless the pupils do something with the information presented in a lecture their learning and retention are liable to be rather than.

Good lecture requires more preparation than most teachers have time to give them. Very seldom can be teacher lecture effectively on the spur of the moment. If a teacher wants a high degree of the effectiveness to his lecture, he must plan them very well. Not only must he plan what he wishes to say but he should also plan what he intends to say.

4.2.1 Advantages of Lecture Method

- ❖ This method is economical because no lab - rotary, apparatus aids etc. are required.
- ❖ A single teacher can teach a large number of students at a time which is impossible by any other method.
- ❖ It is a good method of imparting voluminous information in comparison to other methods of teaching. This is so because much information can be supplied to the students in a short time.
- ❖ It is useful in imparting factual information in relation to some of the thrilling historical and biological incidents in science. Factual information can be five in an effective and interesting way by this method, provided the teacher has a good personality and his manners and manner of appeal are good and the uses the method with wit humor and interesting anecdotes.
- ❖ Listening to lectures given an opportunity to the pupil to have same experience and be attentive. A well prepared and well delivered lecture saves the time of the pupils.

4.2.2 Disadvantages of Lecture Method

- ❖ This method is against the principle of learning by doing. The students are not involved in the teaching learning process at all in his method.

- ❖ The children are not active participants in the learning process but only passive listeners which is psychological.
- ❖ The organisation of the material may be incomplete, the teacher may lay under emphasis on detail, the language used may be very dry in presentation without humor. Then the lecturer fails to make his impression on his pupils.
- ❖ The lecture method is a taxing method on the part of the students. This method gives very little chance of testing the previous knowledge of the pupils.
- ❖ The capacity to understand various forms of student to student as the class is a heterogeneous group of students. There is no assurance whether the students understand all that the teacher is saying.
- ❖ The rate of imparting information by the teacher may at times be too rapid for the learner to get necessary connection of the thought.

4.2.3 Application:

In the pure form, the lecture method is unsuited for the middle and high school classes. But it can be adapted to suit our needs but punctuating by giving opportunities to pupils to answer them and ask questions frequently and clear doubts and by using visual aids and blackboard freely.

In this adopted form, the lecture method has a distinct place in the teaching of Mathematics.

- ❖ Whenever we introduce a new topic or new specimen it has a definite place. For example wild animals, water plants etc., but such talks should always be brief and richly illustrated so that interest is created even at the start.
- ❖ Whenever we conduct a series of experiments under the same topic e.g. Photosynthesis, transpiration etc. they may remain as. Children's incomplete knowledge, which should be filled in by the teacher's informal talk.
- ❖ The most valuable use of this method is to encourage the pupils in learning more about the lives of great pioneers of science like Louis Pasteur, Robert Koch, Ronald Ross, Charles Darwin etc. such type of talks have a very definite place in science.
- ❖ It is however suggested that along with the lecture method practicing of note taking should be given to students of high and higher secondary stage.

- ❖ Lecture should be followed by written tests to measure the success of the lecture. At the end of the lecture opportunity should be provided to students to ask questions and to answer the question and to make sure that salient points have been understood by the students.

You know that lecture method has immediate advantages. So the lecture has strong influence on teaching method. The students have to listen quietly. Take notes and assimilate information within the stipulated time. But the learning process is not simple. The 'pouring in' method (lecture) is not recommended at secondary level. Teacher should limit themselves to briefly presentation such short description of the process, statement of fact etc. any oral presentation is effective only to the degree with the imagines are entered. That is why slides, exhibits and demonstration materials are helpful to the speech.

4.2.4 How to make Lecture Effective:

Let us see how to make lecture effective.

- ❖ Content should be presented systematically.
- ❖ Objectives should be stated clearly.
- ❖ The outline of the content should be presented.
- ❖ Pertinent illustration should be used.
- ❖ There should be element of humor.
- ❖ Experience should be shared.
- ❖ Summarization should be carried out then and there.
- ❖ Teacher's personality should be borne in mind.
- ❖ Speed must be appropriate.
- ❖ Voice should be audible, clear, impressive and modulated.
- ❖ Pronunciation must be approachable.
- ❖ Pupils should be encouraged to take notes during lecture using following methods.
 - a) Outline method.
 - b) Salient point method.
 - c) Verbatim method.
 - d) Short hand method
- ❖ Teacher should possess extensive reading habit.
- ❖ Appropriate teaching aids should be used appropriately.
- ❖ Interaction should be maintained by questioning the pupils and clarifying their doubts.
- ❖ Dictating notes should be avoided.

4.3 LECTURE - DEMONSTRATION METHOD

The demonstration helps the teacher to show and illustrate an object or technique to the class. The specific purpose of the demonstration is to set forth the relationships in correct form at as they occur. Demonstration is an important step in skill lessons. It is true that a method all the pupils do not use their but the teacher reinforces the learning but the use of two channels the ear and eyes. The spoken words are link between series of action by the teacher. Demonstration is the link between explanation and practice.

Demonstrations are not useful teaching devices for the biology teacher. Demonstration serves the purpose of the lecture method and the laboratory method. So demonstration should be used frequently and properly.

The word 'demonstration' means showing. So when the teacher shows experiments it is a demonstration.

a) Steps involved in lecture cum demonstration

4.3.1 Planning demonstration

- ❖ An effective demonstration requires purposeful and careful planning. Attention must be given to many factors like visibility, audibility, attention, audience participation.
- ❖ First the teacher has to consider whether demonstration method is proper method and whether the materials to be used in demonstration will be visible to the students.
- ❖ To ensure the visibility the simple and large scale should be used. The background behind the apparatus should be proper, like cardboard, clothes etc. proper highlighting is essential and backlighting will be better. One or two spotlight may focus on demonstration apparatus. Demonstration materials need to be always on the table. The materials may be kept inside the box or covered by cardboard.
- ❖ The teacher prepares sufficiently earlier to the time of demonstration and rehearses less before demonstration.

4.4 TEAM TEACHING

Teacher teaching is a teaching technique that involves a group of teachers who are jointly responsible for the planning and instruction of a course. The group of teachers may be as few as two or as many five or six depending upon the nature and objectives of the course, the size of the class and the facilities to be used. The teacher of the team cooperatively develop a programme of

instruction and share in teaching, evaluation and course improvement.

Team teaching is effort to improve instruction by recognition of teaching personnel, involving the assignment of two or more teachers ranging from specialist to observer. Team teaching eliminates the rigid grouping based on one or two criteria and allows for variations in student grouping depending upon the outcomes being sought. Evaluation of teachers is based upon the common observation of several teachers. It further encourages teachers to become specialist in one or more areas.

Team teaching is a form of teaching two or more teachers form different disciplines share the responsibilities for the planning, presentation and evaluation of lessons prepared for two or more classes of students. In the method of teaching the whole of the participants working together will make a greater contribution than the sum of the individuals working alone.

Team teaching composed of two or more teachers goes hand in hand with instructional improvement through:

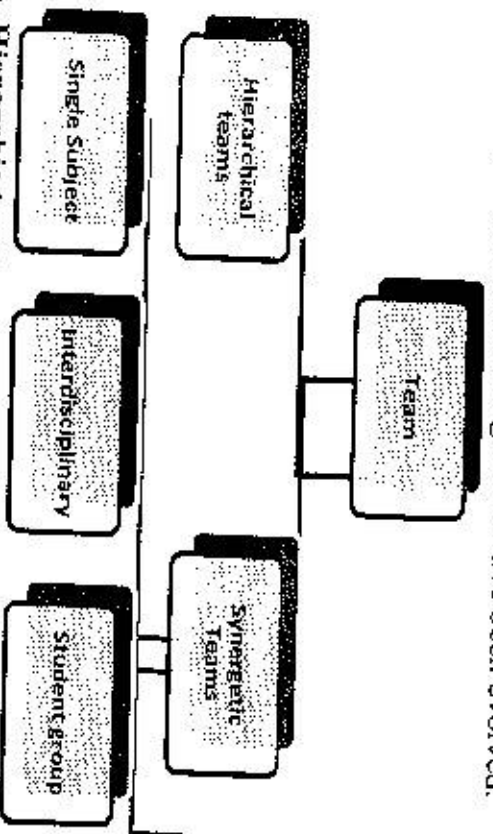
- ❖ Better utilization of staff.
- ❖ Greater flexibility in grouping, scheduling, and the use of the space.
- ❖ Provision for large group, small group and individualizes instruction.
- ❖ Increased use of audio visual aids.

4.4.1 Characteristics of Team Teaching

- ❖ As a method of organizing instruction team, teaching efforts an opportunity to incorporate new innovations.
- ❖ It also provides for continuous improvement of instruction.
- ❖ Because of the flexibility in grouping students, it can also effectively provide for individualized instruction.
- ❖ The staff as a team, plan, conduct and evaluate the programme co-operatively.
- ❖ The teaching provides the students with instruction from the specialist rather than the generalist.
- ❖ The concept of team teaching has been formed to achieve certain educational goals such as providing a means of continuous improvement of the instructional program providing more knowledgeable teachers and developing greater instruction.
- ❖ Effective use of audio visual aids.

4.4.2 Types of team teaching:

Two major types of teaching teams have been evolved.



A) Hierarchic teams:

This type of team can be compared to a pyramid, with the team teacher at the apex, the master teacher just below and the regular teacher and aids still below. A major purpose of hierarchic to provide the teachers with a mean of professional advancement. The advantage on establishing level tends to diminish teacher's role.

B) Synergetic team:

Synergetic team are formed by two or more teachers willing to cooperate as professional equals. Aims may be developed to work within conventional facilities and schedules. In a synergetic team the instructional leadership rotates according to the need. These teams are also co-operative or associated team. Here several teachers plan and carryout instructional program on a co-operative basis with no specific ranks officially assigned for the staff members.

However a co-operative unit may elect a team coordinator to assume responsibility for scheduling team meetings, maintaining record facilitate communication and maintaining contact with administration.

1) Single subject teams:

In these two or more teachers agree to teach the same subject at a particular grade level to the same group in a common period.

If two teachers constitute a team the student group might number fifty to sixty.

2) Interdisciplinary teams:

In this type of team the teachers of different subjects agree to assume responsibility for the same large group of students and are given in a block time in which to work with team. Thus the block of time for mathematics, English, social studies and science team might extend to four periods daily and the student group might consists 120 or more students.

3) Student group teams:

In some situations of a team 5-7 teacher assumes full academic responsibility for the entire school day for a group of student. This team can function almost as a separate entity. Fieldtrip, movies and guest speakers can be scheduled without affecting the rest of the school. Community study and projects can be fitted in without disturbing other classes.

4) Hierarchical teams:

In this type team one professional teacher is designated as team leader and functions as administrator in charge, the remaining teachers perform professional duties as agree upon by the team.

1. Remove clerical drudgery from the professional act of teaching.
2. Permit highly paid, highly trained, highly qualified teachers to focus upon the activities that are genuine to their areas of expertise.
3. Provide a mechanism for introducing competent non-professional into school when their knowledge and experience can be of use.
4. Establish an atmosphere for growth, development of new teachers.
5. Make maximum use of the materials and resources accumulated by effective teachers over the years, and serve to motivate beginners to augment the collection to meet their current needs.

Initiating team teaching

Team teaching covers as a large number of instructional arrangements. The teacher should fulfil the 3C's of team teaching that is they do communicate, co-operate and collaborate. The team

teaching should be planned according to available facilities in the school.

4.4.3 Advantages of team teaching

Team teaching has several basic advantages over the conventional teaching using individual's class teachers.

- ❖ Team teaching provides opportunity to redefine goals of education, to organize instructional patterns and in-service instructions
- ❖ Individual student learns difficulty in group of varying sizes and team teaching which usually provide
- ❖ Provides for instructions in both large and small groups can meet these individual differences better than the usual classroom teaching.
- ❖ The large group instruction in team teaching gives both student and teachers a welcome break from the monotony of conventional class room instruction.
- ❖ Large group sessions enable more students to benefit from instruction by the most skilled teachers.
- ❖ Under team teaching students can beneficially be exposed to teachers with varying backgrounds and different areas of specialization.
- ❖ When non-certified personnel and clerical aids are part of the team, the experienced teachers can be freed of non-professional time consuming tasks.
- ❖ Students are given more chance of self-directed study and independent research.
- ❖ The content of different subjects may be correlated more effectively in team programme.
- ❖ Team teaching enables teachers to share information and ideas which help them solving problem and improve their background.

4.4.4 Disadvantages of team teaching:

The following are some of the disadvantages against of the team teaching

1. Team teaching demands more staff.
2. Very large groups are difficult to control and evaluate.
3. Opportunities for the pupil leadership may be lost because of the complexities of the program and size of the group.
4. Students may find it difficult to establish relationship with different teachers teaching with the same effect.

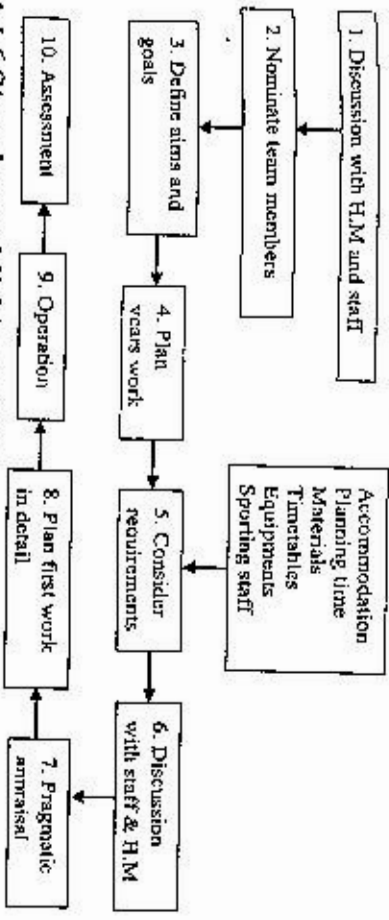
5. Personality differences as well as those of the age, training and experience make some adverse effects.
6. Team teaching acquires so much of complicated administration, extra work and additional expenditure.

4.4.5 Implementing team teaching

Choosing a team:

There is no single approved size of teams. They generally range from two teachers for about 50 students to three or four teachers of about 75,100 students. The size of the team depends on the available space personnel and specific number of children involved.

D) Application of team teaching to the teaching of Commerce and Accountancy



4.4.6 Steps in establishing a team Implementing team teaching

Choosing a team:

There is no single approved size of teams. The generally range from two teachers for about 50 students to three or four teachers about 75,100 students. The size of the team depends on the available space personnel and the specific number of children involved.

Differentiating staffing:

The team teaching staff is composed of professional, and non-professional. The non-professional includes paraprofessional and clerical aids. One the teacher is assigned as the team leader.

Planning a team:

It is essential for the successful functioning of team teaching. We must also decide in advance on their goals, the allocation of

responsibilities, record keeping, and methods of working together, arrangement of space, and materials and resources to be ordered.

Organizing the space:

When one or more teachers have decided to work co-operatively they may require more space. Available facilities like library, common halls classrooms etc can be used.

Organizing the students: The student in the team a divided into small homogeneous groups for better administration and care. The group is based on the age and ability of the students.

Organizing the work:

Teacher must decide on the division of responsibilities in their team. In organizing the work the teachers are given specific responsibilities according to their ability, skill and interest.

Planning the work:

Adequate preparation time is essential for the success of teaching. Planning sessions include a review of the current work for the class. Teachers should make short term and long term plans for future work. They should reveal individual students work. Each child should be discussed by all members of the team at least once a month. A briefly review of the records is usually helpful.

Records of the team:

Record of student work in team is important, they are necessary for the team member to plan his work with the student. The team member should assume responsibility for checking all the records of a particular student regularly.

Meeting the parents

Parents meeting should be arranged at regular intervals and the student's problems and progress are discussed with them. This helps to stimulate a better understanding between the student and the teacher and parents.

The student teacher:

If a student teacher assigned to work with a team for period of this practice he will be able to observe a Variety of teacher, methods and possibly subjects with in the same context, he will join in and suggestion to draw in the preparation of his own lessons, at the same time having a cross section of opinions and advice on his teaching techniques. He will be given a greater intensity of teaching experience. this includes the parallel teaching to two

large group, one thought by the student teacher other thought by an experience staff for the purpose of direct comparison, demonstration and conclusion.

The team concept can offer more than the promise of a better education for the students today and tomorrow. Even this promise will fade unless the teacher and school staff

- ❖ Know how to utilize the team to facilitate the planning and organizing of instruction.
- ❖ Understand the purpose and processes of team operation.
- ❖ Help each child get opportunity to learn what he needs to learn in a way he is able to learn it.

4.5 LEARNER CENTERED INSTRUCTION - SELF LEARNING

Self-learning is taking in information, processing it, and retaining it without the need for another individual to be teaching it in order for the understanding to occur.

Simply stated, self-teaching (or self-learning) requires the ability for a student to work independently. Self-taught students are motivated by the sense of a job well done. They are self-motivated because they have the confidence that if they don't know an answer to a question, they know how to use the resources available to find out the answer.

Self-taught students often can complete course work in a fraction of the time it would take to finish in a teacher-directed setting. Self-teaching should be ingraining in the student a concept of continual success, gradual and steady success that results from diligence and the pursuit of excellence.

"In its broadest meaning, 'self-directed learning' describes a process by which individuals take the initiative, with or without the assistance of others, in diagnosing their learning needs, formulating learning goals, identify human and material resources for learning, choosing and implement appropriate learning strategies, and evaluating learning outcomes." Knowles

4.5.1 Importance of self-learning

The importance of self-learning has been pinpointed as follows-

- ❖ Self-learning helps a person in understanding the basic concept of learning and it says that everyone has to learn by himself at the end of the day. As you go deeper you start exploring new horizons that might have been a taboo in traditional form of learning and this can open further doors.

Actively pursuing your goals single-handedly gives the learner the necessary confidence to deal with the eventualities of life by any means.

- ❖ The importance of self-learning is that it takes the crutches away. You are no longer dependent on others and this proves beneficial in the long run. Identifying and tackling problems works as a booster and gives you immunity against issues and problems that a person faces in life.
- ❖ Self-learning helps you to find the ground under your feet. It prepares you for the future where you have to work alone to reach your goal. The resilience and the immunity that you gain here are the key tools that will take you further in life.

4.5.2 Benefits of Self-Learning

- ❖ Student becomes an independent thinker.
- ❖ Student learns to accept responsibility.
- ❖ Student gains the freedom to learn without restrictions.
- ❖ Student earns accountability.
- ❖ Intrinsic rewards become the focus that good feeling inside that comes from a job well done.
- ❖ Student tests well because he is used to tackling problems on his own, which equals confidence.
- ❖ Students retain more naturally when they do the work versus parent's spoon feeding the information into them.
- ❖ Students learn where to go when help is needed. There is no need to worry about gaps in their education because if they need to know something down the road, they will just look it up on their own.
- ❖ Students become more than prepared for college study, which will require motivation and planning ahead.
- ❖ Self-learning gives the opportunity to develop a good work ethic.
- ❖ Self-learning allows the learner to go as deeply into a subject and interact with the subject matter as deeply as he would like to go.
- ❖ Self-learning enables the learner to limit the number of interests undertaken so as not to be spread too thinly.
- ❖ Self-learning allows the family to function as a family without emulating an institution at home.
- ❖ Self-learning eliminates all excuses for not reaching one's potential. It will never be anyone else's fault if the student doesn't learn.

- ❖ Self-learning is more fun than being taught at.
- ❖ Self-learning means that mom can read great books rather than teacher's manuals and text books.
- ❖ Self-learning is the wave of the future now that so much information is available at our fingertips.
- ❖ Self-learning means that babies and toddlers get more attention from mom because she is not busy playing teacher.

4.5.3 Disadvantages of self-learning

The disadvantages of self-learning are as follows:-

- ❖ You do not have readymade material at your disposal and whatever you need will have to be acquired through additional effort on your part
- ❖ You will have to verify your study materials whereas in traditional learning method these were already verified beforehand
- ❖ Working with others help in brainstorming which is not possible during self-learning
- ❖ Teamwork encourages open communication that is not seen during the process of self-learning
- ❖ There is no face-to-face interaction during self-learning
- ❖ The lack of transformational power is a serious limitation of self-learning
- ❖ There is a slow evolution
- ❖ Understanding becomes difficult without someone to explain things
- ❖ Waste of time as the learner has to spend considerable time in finding viable methods and understanding the material by himself.
- ❖ There is no guarantee that whatever you are learning is correct.

4.6 FORMS OF SELF LEARNING: PROGRAMMED INSTRUCTION/PROGRAMMED LEARNING STAGE

4.6.1 Programmed Instruction: An Important Innovation

Programmed instruction or programmed learning is one of the important innovations in the teaching-learning process. Programmed instruction is a carefully specified, systematically planned, empirically established, skilfully arranged and effectively controlled self-instructional technique for providing individualized

instruction or learning experiences to the learner. The subject matter or learning experience is logically sequenced into small segments. The learning experience is self-corrective. It is an application of the principles of behavioral sciences and technology in the field of education.

The English authors prefer to use the term programmed learning. On the other hand, American writers prefer the use of programmed instruction.

4.6.2 Brief History of Programmed Learning

It is held by some educators that 'Gita' is the first example of programmed learning. It is also maintained by them that the programmed text of the 'Gita' has several ingredients of programming: initial behaviours, small steps, active participation, terminal behaviour, immediate feedback and self-evaluation by the learner.

Several educators regard Socrates as one of the earliest programmers who developed a programme in geometry. This was recorded by Plato in the dialogue *MENON*. Socrates used to guide his followers to knowledge by conducting them conversationally along a path from fact to fact and insight to insight.

4.6.3 Programmed Learning as 'Technological Revolution' In Education

Programmed learning emerged in the beginning of the 20th century from the efforts of American psychologists. E.L. Thorndike (1874-1949) was the first psychologist whose findings bear direct relevance to programming. Other important psychologists who have made significant contribution in the field are Sidney L. Pressy, Robert M. Gagne, Robert Mager and B.F. Skinner.

Programmed learning is related with the 'Law of Effect' as explained by Thorndike. Sidney L. Pressy, a psychologist of Ohio State University, is credited for developing in the middle 1920's practical machines which could teach as well as test. The teaching machines as developed by Pressy present a series of questions to a student and inform him immediately whether his response is right or wrong.

In 1943, Skinner and his two other colleagues started programming by teaching a pigeon to roll a small bowling ball by operant conditioning. By 1954, Skinner and James G. Holland devised the auto-instructional methods which have served the

present generation as the basis for present work in programmed instruction. In Skinnerian programmed instruction whether mechanized or otherwise the learner is initially asked questions which he can easily answer correctly without any previous study of the particular lesson. The learner is taught by the sequence of questions. He is asked more and more as the lesson proceeds in very small steps.

In 1955, Norman A. Crowder developed what he calls "automatic tutoring by intrinsic programming" as against "extrinsic programming" developed by Skinner.

Robert Mager (1958) gave a new concept known as "Learner Controlled Instruction" which is a kind of Socratic dialogue in reverse, in which the learner led the instructor. The instructor remained silent until the learner himself stimulated the instructor with questions that suggested the needed illustrations, demonstrations, practice or some other help.

Stolurou, at Illinois, had been developing a process which should provide for greater individualization by measuring needs and developing programmes that require a computer to assist instruction.

In 1962, T.F. Gilbert gave formalized expression of his technology of education called *Mathetics*. Pennington and Slack expressed in 1962 further detailed methods of preparing lessons from mathetic principles.

E.L. Rothkopf in 1965 suggested a term "mathemagenics." Mathemagenic behavior involves three characteristic activities: (1) The transformation of a nominal stimulus into an effective stimulus (which is to take place within the organism); (2) Some form of elaboration (which, for instance, may take the form of echoic responses); (3) Process exemplified by problem solving and inductive reasoning. These activities must take place before any learning can occur. Rothkopf points out that as the learner reads a text, he experiments with and internalizes various kinds of mathemagenic behaviours. The test questions that learner encounters sort out and extinguish those behaviours which were not effective.

In 1965 also, Lawrence Stolurou brought the idea of Computer Assisted Instruction, (CAI).

4.6.4 Definitions and Meaning of Programmed Instruction

1) Arthur A. Luonsdaine (1964): An instructional programme is a vehicle which generates an essentially reproducible sequence

of instructional events and accepts responsibility for efficiently accomplishing a specified change from a given range of initial competencies or behavioral tendencies to a specified or terminal range of competencies or behavioral tendencies.

2) **Edgar Dale:** Programmed learning is a systematic, step by step, self-instructional programme aimed to ensure the learning of stated behaviour.

3) **Gulati and Gulati:** Programmed learning as popularly understood is a method of giving individual instruction in which the student is active and proceeds at his own pace and is provided with immediate knowledge of results. The teacher is not physically present. The programmer, while developing programmed material has to follow the laws of behaviour and validate his strategy in terms of student learning.

4) **G.O.M. Leith (1966):** Programme is a sequence of small steps of instructional material (called frames), most of which require a response to be made by completing a blank space in a sentence. To ensure that expected responses are given, a system of cueing is used, and each response is verified by the provision of immediate knowledge of results. Such a sequence is intended to be worked at the learner's own pace as individual self-instruction.

4.6.5 Fundamental Idea of Programmed Learning

Edward L Thorndike (1912) described the fundamental idea of programmed self-instructional materials in these words. "If by a miracle of mechanical ingenuity, a book can be so arranged that only to him who had done what was directed on page one, would page two become visible and so on, much that now requires personal instruction could be managed by print." This 'miracle' later on became a reality.

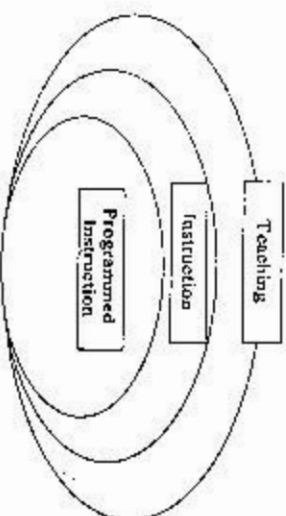
In programmed learning, it is said that the most efficient, pleasant and permanent learning takes place when the student proceeds through a course by a large number of small, easy-to-take steps. Wilbur L. Schramm (1962) lists the essential elements of programmed instruction as:

- An ordered sequence of stimulus items,
- To each of which a student responds in some specific way,
- His responses being reinforced by immediate knowledge of results,
- So that he moves by small steps,
- Therefore, making few errors and practicing mostly correct responses,

(i) From what he knows by a process of successively closer approximation, toward what he is supposed to learn from the programme.

4.6.6 Teaching Instruction and Programmed Instruction.

According to Edgar Dale, "Teaching" is a broad, vague, ill-defined term and instruction is a purposeful, orderly, controlled sequencing of experience to reach a specified goal. Programmed instruction is a sub-head under instruction and represents a more rigorous attempt to develop a mastery, over specified goals to secure insured learning."



Programmed Learning and Programmed Instruction.

These two terms PL and PI have been used interchangeably in educational literature. The English educators prefer the use of programmed learning (PL) and the American writers on the other hand use the term programmed instruction (PI).

Programmed Instruction and Educational Technology.

Programmed instruction is a part and not the whole of educational technology. It is a strategy of teaching and learning and self. Instructional texts or audio-tutors. It may be treated as software approach for instructional technology. Software approach refers to application of teaching learning principles to the deliberate and direct shaping of behaviour. Programmed learning is different from teaching machines and devices usually termed as hardware educational technology.

Essence of a Programme.

The essence of programme lies in writing it in such a way that every student will eventually be able to complete it without a mistake. The key to the programme is to ensure that the student tries the correct answer. The material presented to the student may be in the form of a book or simple machine or a complicated

computer. In every case, the student gives his response and then checks the correctness of the response.

4.6.7 Salient Characteristics of Programmed Learning

- ❖ The objectives underlying the programme are defined in explicit and operational terms. This makes the terminal behaviour desired to be built up through the programme measurable and observable.
- ❖ The subject matter of the programme is presented by breaking into small steps in a logical sequence. The small steps stress the gradual nature of the increase in complexity and the smoothness of the transition from one item to the next. Information grows in depth. Changes occur in quality and quantity.
- ❖ Programmed instruction is a process of constructing sequences of instructional material in a way that the rate and depth of learning are maximized, understanding is fostered and the motivation of the student is enhanced.
- ❖ Assumptions about the learner are clearly stated and put in definite terms in a programmed learning type of situation. These assumptions may relate to the particular level of reading competence of the learner, his command of vocabulary and his back ground in the subject matter.
- ❖ Programmed learner emphasizes the interaction between the learner and the programme.
- ❖ The learner is made to respond actively by asking to fill in the blank, count the number of coins in a row or complete a series of numbers. For doing it correctly the student receives reinforcement and establishes a pattern of stimulus-response interaction.
- ❖ A programmed instructional sequence takes into consideration the initial behaviour of the learner with which it starts and the terminal subject matter competence which the learner is to achieve.
- ❖ In a programmed learning situation, the learner progresses at his own pace.
- ❖ A programmed test provides for immediate feedback information. This is based on the theory of reinforcement

which emphasizes that the learner learns from the sequence of responding and not from the making of response itself.

- ❖ Programmed instruction takes care of the fact that there are even differences in the rate at which an individual learner learns various kinds of subject matter.
- ❖ Programmed learning enhances the capacity of the learner to discriminate or to generalize by frequent application and thus offers the learner an interesting challenging project.
- ❖ In programme learning a learner moves in every specific way as opposed to traditional procedures of teaching where the learner moves in very general way.
- ❖ Programmed learning provides for constant evaluation through the record of learner's responses. The quality of the programme can be improved through checking the number of errors at each step and the learner's progress can be evaluated by looking into various types of responses produced by the learner.
- ❖ Programmed learning is based strictly on the behavioristic principles of psychology. A fair amount of stress is laid on the examination and development of understanding through the handling of various cues in the learning process.

4.6.8 Difference between Programmed Instruction and Traditional Method

Programmed learning may be explained in brief as under:

- (a) The learner is presented with a Stimulus which gives him information, demands on his response, or does both.
 - (b) There is a continual necessity for the learner to utilize his information in making some response.
 - (c) After responding, he is presented with information (feedback), which enables him to ascertain the appropriateness of response.
- Need for Programmed Instruction.** Technology or automation is the only solution to the challenge posed by 'knowledge explosion' and 'population explosion'. The present era, therefore, is passing through a period of a silent revolution in the field of education. Programmed instruction is an innovative step in this direction towards automation and individualization of instruction.

Programmed Instruction	Traditional Method
1. It is an individualized technique of instruction	1. It is a group technique.
2. It is based on the teaching principles that have been known for years.	2. It becomes difficult to apply teaching principles in crowded classrooms.
3. It presents the instructional matter step by step in logical order.	3. It presents the instructional matter as a whole.
4. The size of the unit of information presented to the pupil is a small bit of information.	4. The unit is a lengthy one. There is no provision for response from the students in the form of answers to questions.
5. Immediate feedback is given to the learner	5. The learner does not get immediate feedback.
6. Objectives are defined very clearly in operational terms	6. Objectives are not well-defined and are usually vague.
7. The programmer prepares his programme with care and precision.	7. Little preparation is made.
8. Programme is prepared in such a way that the student automatically participates actively by making responses continually.	8. The student usually remains a passive listener and the teacher himself does the summarizing and reviewing.
9. A programme is developed empirically through a series of tryouts and refined gradually. Effective sequences of frames are retained and ineffective ones discarded	9. It is usually found to be very difficult to modify traditional instruction on the basis of student reaction.

4.6.9 Fundamental Principles of Programmed Learning

Principle of small steps.

It is shown by experiments that even the duller students can learn as effectively as the brightest students if the subject matter is presented to them in suitable small steps. When we divide the task to be learnt into very small steps, and ask the students to learn only one step at a time, then probably all the students will be able to learn one small step at a time and sequentially learn all the steps. It

is a difficult task to climb a mountain but once steps are built even a child can climb the mountain very easily. This is known as the principle of small steps.

Principle of activity responding.

The second psychological principle is that the students learn better and faster when they are actively participating in the teaching-learning process. In our class room teaching the teachers do ask a few questions and the students respond. But it is not possible for the teachers to ask all the students to respond at each small step. A teaching machine text or a programmed text contains a large number of questions-one question at each small step and the students respond actively.

The principle of active responding is used for the programmes. The teaching machines and programmes have proved to be superior because they provide opportunity to every learner to respond at every small step.

Principle of reinforcement.

Every response even approximately correct must be reinforced immediately. Delayed reinforcement fails to work. This is possible only when a teacher has to teach only one student at a time. The most ideal situation is when the teacher can cater to the needs of his students individually. But in classroom teaching this is hardly possible. No teacher, however efficient and sincere he may be, can reinforce each correct response of each of his students as soon as it is made in a classroom situation where he has to teach about 40/50 students. The teaching machines and the programmes do the job more efficiently.

Principle of self-pacing.

The programmed instruction is based on the basic assumption that learning take place effectively if the learner is allowed to learn at his own pace. Therefore, a good programme of the material always takes care of the principle of self-pacing. A learner moves from one frame to another according to his own speed of learning.

Principle of student-evaluation or student testing.

Continuous evaluation of the student and the learning process leads to better teaching-learning. In the programmed instruction, the learner has to leave the record of his responses because he is required to write a response for each frame on response sheet. This detailed record helps in revising the programme. Prof. Edward F.O.

classifies principles of programmed instruction in two broad categories:

4.6.10 Fundamentals of Programmed Instructions

Fundamentals of programmed instruction are given below:

- ❖ Stimulus and Reaction
- ❖ Behaviours and Behaviour Repertoire
- ❖ Reinforcement
- ❖ The Transfer of Stimulus Control
- ❖ Feedback
- ❖ Confirmation
- ❖ Prompting
- ❖ Generalization and Discrimination
- ❖ Gradual Progression
- ❖ Successive Approximation
- ❖ Diagnosis and remediation
- ❖ Retrogressive Chain
- ❖ Learner Controlled Instruction.

These elements are being explained individually as follows

1) Stimulus and Response

Such situations, event or person or change in the environment which bring changes to the student's behaviour are called as Stimulus. Stimulus creates a situation for a specific response. In programmed instruction course contents are presented into small fractions in a logical order. Every fraction acts as a stimulus and these fraction prepares students to responses (by creating appropriate situation). Right stimulus instructs students for the proper and right response and provides new knowledge. Since stimulus and response are useful for change in students behaviour and in some way behaviour.

(1) Behaviour and Behaviour Repertoire

By behaviour, we meant to such activities which are performed by students to attain teaching goals. Behaviour is the total response of the organism to situations of life. It considers inner and overt behaviour and also combines the study of inner mental processes and other outer behaviour in the context of programmed instruction, a group of stimulus-responses is called as behaviour response is a unit of behaviour. Stimulus-response collectively develop behaviours. Behaviour Repertoire is that

series in which many responses are managed in a group by the logical method.

(2) Reinforcement

Reinforcement is that event which occurs after completion of a process and reinforces that process. In order words, the possibility of the occurrence of that gets increased.

Reinforcement is related to such events of the environment which increase the possibility of a response. New behaviours or change is based on such responses which are powered by the stimulus. Such events or situation of stimulus which create responses are called as reinforcement (Sharma, 1966)

(3) The Transfer of Stimulus Control

In the beginning of programmed instruction material, when student responses with Stimulator, he is already familiar to them. As he moves forward, responses help in reaching from the entering behaviour to ending behaviour and stimulus control keeps moving in learning order. This is called as the transfer of Stimulus Control.

(4) Feedback

Feedback is that process in which students are made aware to their weaknesses, faults and errors so that students can improve, also student's good features, good work, their quality and strengths are explained in this process so that they can display them even further into their behaviours. Reinforcement increases the possibility of response while feedback is a powerful tool for change in behaviours. Feedback methods improves the student's behaviours, develop them and make desired changes in them.

(5) Confirmation

Confirmation is also called as the third principle of programmed instruction. The feedback is immediately provided that students' response is correct due to which students move forward. Confirmation is a form of feedback due to which students attain new knowledge and also reinforcement is provided to them. Student gets the completeness of teaching material by moving forward through ordered fractions based on confirmation of his response.

(6) Prompting

In programmed instruction student have to response for every fraction for which an extra stimulus is used. This is

called a Prompt. Any information contained in a frame to help the learner to respond correctly is known as prompt or cue. This will prevent students from making the wrong response.

(7) Generalisation and Discrimination

The ability to acquire the skills, aptitude and knowledge etc. In a situation and to response those in a similar situation for the same elements is called generalization.

(8) Gradual Progression

In programmed instruction, students are slowly moved from a chain of entering behaviour to ending behaviour through gradual progression. In gradual progression, this matter is taken care of. Student develops complex behaviour through slow response. Each fraction of the content has an arrangement that which students will gradually move up on the path of progress by relating the student's prior responses to his further responses.

(9) Successive Approximation

In programmed instruction material the prior response of learner are reinforced. The processes required to reach the ending behaviour are divided into small fraction in according to the logical sequenced method. Learner's responses reach near the ending behaviour by reinforcing at every fraction on the basis of successive approximation.

(10) Diagnosis to provide remedial instruction

Diagnosis and Remediation refers to provide remedial instruction on students according to their needs, weaknesses and difficulties by diagnosis their difficulties and weaknesses. Student's remediation should be done on the basis of their diversity. When student makes incorrect response then his difficulty, fault or weakness gets noticed for which he get remedial instruction on wrong-page and he receives instruction from material to improve his fault.

(11) Retrogressive Chain

To reach the level of mastery, a progressive chain is followed in the linear programmed instruction but T.F. Gilbert has used retrogressive chain in his instruction. Opposite to progressive chain, this chain is started from the endpoint and is ended at the starting of the chain such as reverse counting or to learn reverse multiplication table (from 100 to 1). This chain is more useful in math.

(12) Learner Controlled Instruction

This concept is the contribution of Robert Mager. The importance is given to students in this instruction. For preparing programmed instruction material, student-centered instruction sequence is more effective than any other instruction sequence. So it should be used at the beginning of preparing instructional material, then we should move further by establishing relationships with the objectives. It helps making instruction student-centered.

4.6.11 Major Fundamental Principles of Programmed learning or instruction are the following:

1. Principles of small steps
2. Principal of active responding
3. Principal of immediate reinforcement
4. Principal of self-pacing
5. Principal of knowledge and progress of student testing

4.6.12 Mandatory and Optional Principles

Mandatory Principles

- These include:
1. Principle of objective specification.
 2. Principle of empirical testing.
- It includes: (a) Individual try-out (b) Small group try-out (c) Field try-out.

3. Self-pacing.

OPTIONAL PRINCIPLES

4. Overt responding.
5. Immediate feedback
6. Small step size.

4.6.13 Rule for the art of Programming

Klaus has summed up 12 rules of the art of programming and building instructional frames:

- ❖ **Active responding.** A learner learns from making a response and not from hearing or seeing it.
- ❖ **Proper cueing.** The provision of proper cueing is the main difference between a test question and an auto-instructional frame
- ❖ **Appropriate context.** This is the kind of stimuli that will be capable of evoking the desired response at some time in future.

- ❖ **Small steps.** It is very desirable that there should be as many small steps as possible.
- ❖ **Sequence of topics.** The topics of the subject-matter should be carefully sequenced so as to provide adequate motivation.
- ❖ **Frequent repetition.** Cues associated should be changed and frequent repetition provided in frames.
- ❖ **Knowledge of subject-matter and technical accuracy.** A sound knowledge of these is very necessary for preparing a suitable and effective programme.
- ❖ **Lecturing to be avoided.** The learner should be helped to learn and this is possible in case we provide facts to the learner.
- ❖ **Evoking a relative response.** Right kind of illustration should be used for evoking a response.
- ❖ **Providing cues in adequate number:** Excess of cues will prevent the learner from making his own generalizations and discovering principles.
- ❖ **Not assuming too much knowledge.** The programmer should not assume too much knowledge on the part of the learner.
- ❖ **Presenting of facts in frames.** Only one fact should be presented in one frame.

4.6.14 Effects of Programmed Learning on the Teaching Process

- ❖ Tutorial experiences are provided for individual learners on a large scale whereby the learner may proceed at his own rate.
- ❖ Programmed instruction can exercise control on homework and individual study.
- ❖ Programmed instruction can enable the learner to catch up the loss of absence from school.
- ❖ Programmed instruction provides a technological solution to the problem of individual differences. Programmed instructional materials enable the teacher to gear up his presentation to what he senses is the average, too slow for the bright student and too fast for the slow one.
- ❖ The availability of instructional material of a programme and its powerful tools make the teacher much more a professional man than he now is.
- ❖ By interpreting the feedback obtained through oral questions and discussions in the class, the teacher can intelligently vary the flow of information or further instruction.

- ❖ A basic feature of teaching machines is 'immediate knowledge of results' followed by immediate correction of errors and rewards for correct answers. It has been established by research that more frequent the rewards, the better the learner assimilates the material.
- ❖ It eliminates cheating because the learner has to gain nothing by it.
- ❖ The students like the use of machines than studying a textbook.
- ❖ It eliminates the influence of teacher variable in research.
- ❖ Immediate scoring promotes the learning process.
- ❖ The technique is valuable in giving tests which can automatically be scored.

Professor John Blyth and John Jacobson describe the following uses of programming:

- ❖ No class time is wasted on drill.
- ❖ Examination of student's work on the programme can enable us to prepare for a class period with prior knowledge of the points that is needed for further clarification.
- ❖ No time is wasted to locate and correct misconceptions the students may have through unchecked practices.
- ❖ Classroom efficiency may increase by about 30 per cent. More material can be covered in less time.
- ❖ Individual differences may be reduced to a considerable extent.
- ❖ Difficulties of the students can be easily diagnosed.
- ❖ Individual and class differences can be easily located.
- ❖ There can be a great increase in interest and improvement in morals.

4.6.15 Merits and Advantages of Programmed Learning

Programmed learning is considered to be a more efficient method of teaching-learning. Its superiority is well-established over the traditional methods of teaching-learning. It is increasingly being used in the advanced countries of the world. It is realized that the programmed instruction has potentialities to revolutionize the theory and practice of teaching. Following are the main merits of programmed learning.

- ❖ A well-programmed instruction is a great thrust in the direction of individualized instruction, as it is tailed to the needs of the individual learner in the class.

- ❖ It permits individual learner to progress at his own speed. An intelligent learner needs no longer to be bored allowed to lose interest on account of his slow progress of other learners of the class. He can make progress as he is able.
- ❖ Since a programme requires continuous response from the learner, it overcomes the inertia and passivity on the part of the learner.
- ❖ The teacher can give explanation in the classroom if the error is common or he may arrange individual conferences on specific points.
- ❖ A well-programmed instruction is a great thrust in the direction of individualized instruction as it is tailored to cater to the needs of the individual students of the class.
- ❖ Learning material in a programmed instruction is presented in such a way that learning becomes an interesting game and the learner is motivated to meet the challenges set by his own capabilities.
- ❖ Programmes are developed by experts. They are empirically tested and modified till they are standardized. A number of learners can use a single good programme and thus save textbooks.
- ❖ In programmed instruction the learner is immediately reinforced to correct his response and this reinforcement sustains the motivation of the learner.
- ❖ The self-instructional technique presents material in which its complexity is simplified through the analysis of the subject matter into small and more easily assimilated segments of information.
- ❖ The introduction of programmed instruction is of great significance for developing countries which are set on the path of educating millions of learners and are short of teachers.
- ❖ Good teachers are freed from the boredom of routine class room teaching and they are in a position to devote more time to more creative activities.
- ❖ The programmed instruction has been used more successfully in teaching the discernment of the logic of various disciplines and inspiring students to creative thinking and judgment.
- ❖ Certain motor skills and intellectual abilities normally taught by frequent drills and rote memorization can be very efficiently taught by self-instructional devices.

- ❖ Self-instructional materials have been found to be very useful in the West in revolutionizing the social setting of the classroom. Problems of discipline have been solved and a new hope for eliminating emotional and social problems has been generated.

- ❖ Programmed instruction enables the teacher to diagnose the problems of the individual learner.

- ❖ The introduction of programmed instruction is very helpful in certain situations where human instructors are not easily available in the required number, for instance small schools in the isolated or hill areas.

4.6.16 Limitations of Programmed Materials

Programmed materials have been severely criticized as a threat to replacing the teacher.

It is also argued that there is too much emphasis in learning facts and very little emphasis on the mastery of principles and concepts.

Some critics of programmed instruction maintain that the user of a programme does not know where he is headed to. They also point out that the learners are not aware of the organisation and programmed instruction is unrelated to other aspects of instruction.

It is also argued that the programmed instruction material is very costly and only rich nations can afford it.

It is also stated that the development and use of programmed instructional material require expert knowledge and training. An average teacher finds it very difficult to make use of this device.

4.6.17 Programmed Learning Approach in Normal Class-Room Teaching

The programmed learning approach can be adopted in normal class-room teaching in the following ways:

- ❖ A teacher can make use of the principles of programmed learning such as active responding, minimal errors and confirmation while teaching various subjects in the conventional manner.
- ❖ A teacher can define behavioural objectives in advance of teaching.

- ❖ A teacher can validate the instructional systems of a class in terms of the performance of learners immediately after teaching is over.
 - ❖ A teacher can regulate questions and answers. The answer of a learner can be immediately reinforced by informing or telling whether it is correct or incorrect.
 - ❖ A teacher can plan the entire instructional programme of a class-room and can treat the terminal behaviour, the pre-requisite skills and content analysis in advance.
- Komoski (1960) an expert has observed "Two thousand years ago the world's first public administrator, a gentleman by the name Quintilian wrote what might be called a handbook for teachers." In it he has one bit of advice which will serve as an excellent starting point for a discussion of programmed learning and its potential uses. His advice is: "Do not neglect the individual student. He should be questioned and praised."

4.6.18 Application of Programmed Instruction in India

Programmed instruction is still in its infancy in India.

Programmed instruction as an optional or elective paper has been included at the B.Ed/M.Ed. level in a few universities in India. It also form a part of the paper of Educational Technology/ Educational Innovation. However, as regards its classroom use, it may be observed that it is almost nil. As far back as in 1966, the Kothari Commission suggested to develop programmed material in different subjects to test the suitability of the technique in Indian conditions. An Association of Programmed Instruction has been formed to coordinate the research being done at different centers in the country. The association also disseminates the information on new studies through its journal issued from time to time. The National Council of Educational Research and Technology has also done some work in the field. In spite of all these efforts, it may be stated that the application of programmed instruction has not yet made an appreciable impact on our classroom teaching. Our methods of teaching still remain traditional, by and large.

Following are the important factors which stand in the way of introducing programmed instruction in Indian schools:

1. Resistance to change.
2. Lack of good programmes and programmers.
3. Lack of facilities

4.7 TYPES OF PROGRAMMING

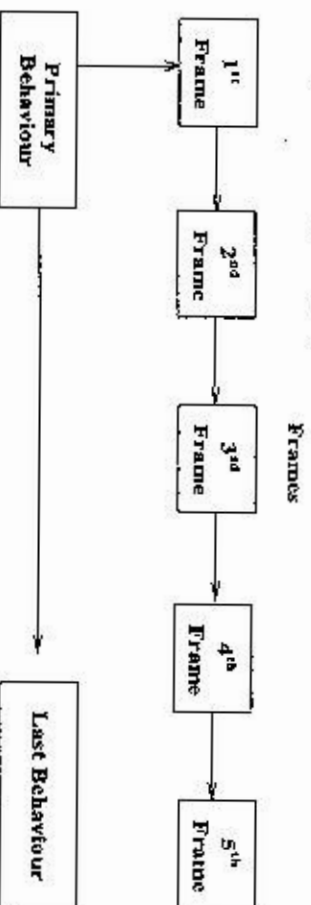
As a result of experimental studies and research, following types of programmed instruction have emerged.

1. Linear or Extrinsic Programming.
2. Branching or Intrinsic Programming.
3. Mathetics Programming.
4. Rules System of Programming.
5. Computer Assisted Instruction.
6. Learner Controlled Instruction.

The first three styles-linear, branching and mathetics are the basic formats. The rules system represents the deductive and inductive approach to teaching. The other two types, Computer Assisted Instruction (CAI) and Learner Controlled Instruction (LCI) are not the basic format of Programming. They are, in fact, the ways and means of providing instruction. Here we have taken up only the basic type of programming.

4.7.1 Linear Programming

B.F. Skinner is the originator of linear programming. It is also called a single track programme. According to Skinner, a creature, a bird or a human being can be led to a desired behaviour by means of a carefully constructed programme consisting of small steps leading logically through the subject matter from topic to topic, provided each step is reinforced by some kind of favorable experience or reward. The increments in information which the learner is expected to absorb are small. The favorable experience, or response increases the probability of the same response to occur again in the future. The process of rewarding the correct response to a stimulus increases the general tendency to give a response. The process may be graphically illustrated as under:



As shown in the figure, the sequence of frames and path of learning in the programmed learning is systematic and linear. That is why, this type of programming is referred to as linear programming. Hence all the learners have to proceed through the same frames and in the same order.

In a linear programme, learner's responses are controlled externally by the programmer sitting at a distant place. Hence linear programming is also termed as extrinsic programming. In branching programming, learner's response is controlled by learner him. Self internally. It is, therefore, also called intrinsic programming.

PRINCIPLES OF LINEAR PROGRAMMING

1. Principle of small steps.
2. Principle of active responding.
3. Principle of minimum errors.
4. Principle of knowledge of results

CHIEF CHARACTERISTICS OF LINEAR PROGRAMMING

1. A linear programme is a single track or a straight line programme.
2. In this programme, learning material is presented into a series of small steps (frames).
3. Every learner follows the same path in a linear programme.
4. In linear programming, the content is broken into small units ranging from just one word to as many as 40 or 50 in length.
5. In a linear programme, the learner is given a small programme, the learner is given a small amount of information.
6. The sequence of steps remains unchanged.
7. The learner is expected to compose his own answer to each question.
8. The learner is expected to respond actively to each step or
9. The responses of the learner get immediate reinforcement.
10. Linear Programming provides for self-pacing i.e. one can learn according to one's own speed.
11. Linear programming moves slowly but steadily in leading a learner from initial to terminal behaviours.
12. Each of the unit constitutes a frame.
13. In a linear programme, the programmer controls the response of the learner.

14. In a linear programme the learns by avoiding the learners error.

MERITS OF LINEAR PROGRAMMING

1. Immediate knowledge of results acts as a great motivator and releases anxiety and tension.
2. The smallness of the frames brings the sub-goals within the reach of the learner and thereby facilitates secondary reinforcement.
3. Repetition strengthens the responses and ensures learning.
4. Easy nature of the programme provides 'success experience to the learner.

LIMITATIONS OF LINEAR PROGRAMMING

1. In linear programming, the learning process becomes quite dull on account of the following reasons (a) Subject matter is broken into very small pieces (b) Responding is quite mechanical and restrictive and (c) The learning process is quite slow.
2. The use of linear programmer is limited to some subjects and topics.
3. Linear programming cramps the imagination of the learner and initiative for creative, integrative and judgment learning.
4. Linear programming encourages guessing.
5. Linear programming does not develop the discriminating power of the students.

4.7.2 Branching or Intrinsic Programming

Branching or intrinsic programming was developed by Norman A. Crowder (1954) an American technician. According to Crowder, branching or intrinsic programme is one which adopts to the needs of the learners without the medium of any extrinsic device such as a computer. It is not controlled extrinsically by the programmer.

Norman A. Crowder was a technician who was working in the United States Air Force. He was faced with the problem of efficiency of vocational training. His programme is based on intuition. His approach at the most is practical. This type of programme employs multiple choice response patterns. The learner is required to select one right answer out of several responses presented to him.

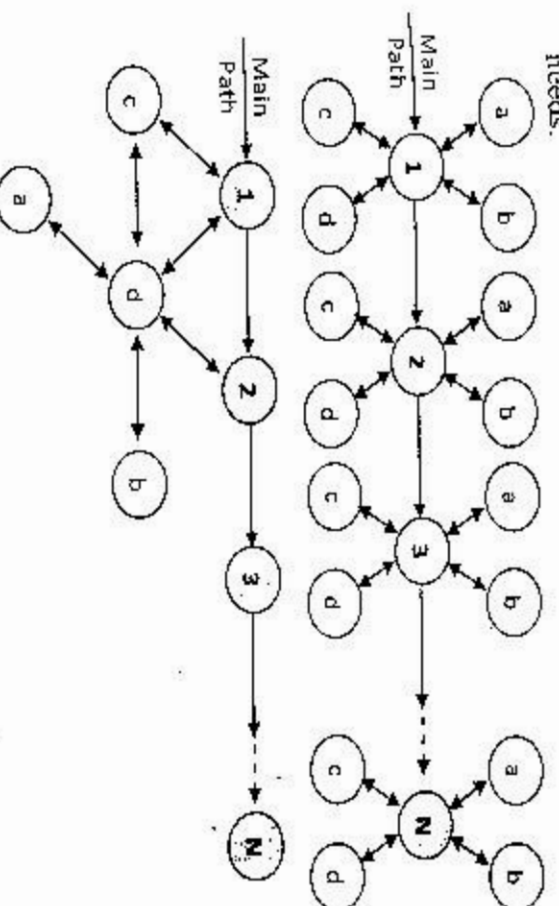
CHARACTERISTICS OF A BRANCHING PROGRAMME

1. A frame may contain two or three related ideas or related sequences.
2. Each frame is of relatively bigger size as compared with the linear type.
3. With a view to answering the question, the learner has to arrive at an answer by setting the relationship between one idea and the other and by filling up the gaps not fully covered in the frame. Only then he can select the correct alternative from the multiple choice.
4. The learner moves forward if his responses are correct but is diverted or branched to one or more remedial frames if he does not give the correct response.
5. Frames explain the matter afresh when response is not correct. The frames ask the learner questions to elicit the right answer and reveal his previous mistakes. The learner, then, returns to the original frame.
6. The cycle goes on till the learner passes through the whole instructional material at his own pace.
7. Branching programme is a choice of items where the learner selects the correct answer whereas in linear programmes or Sinner's programme, the learner is constructed rather than selected.
8. In a branching programme, the learner does not follow the frame route. Instead, the learner chooses the response made by the learner. Thus, the learner chooses according to their responses.

ASSUMPTIONS OF BRANCHING

1. Learning takes place better, if the subject-matter is presented in its totality.
2. Learning takes place better, if the subject-matter is presented in the form of meaningful components or units.
3. Wrong answers do not necessarily hinder the learning of a correct answer.
4. Multiple choice items help more in the learning process.
5. It is based on the possibility of detecting and correcting errors.
6. Basic learning takes place during the learner's exposure to the new material.

7. Learning takes place better if the learner is allowed sufficient freedom to take decisions for adopting the instruction to his needs.

**Main Path and Branching in Branching Programme****MERITS AND ADVANTAGES OF A BRANCHING PROGRAMME**

1. Big size of a frame as well as the branching minimizes unnecessary repetitions and responding, thus reducing the amount of learning time and fatigue.
2. The pitfalls and consequences of erroneous logic are usually explained in the remedial frames so that the learner not only gets the correct response but also understands why some other response is not correct.
3. Instead of simple response it provides alternatives in the form of multiple choice.
4. Through its broad frames, branching programme provides for more freedom to respond and scope of choosing one's own path of learning according to one's need. Thus it helps in maintaining the interest and initiative of the learner.
5. Branching programme is helpful in the development of the power of discrimination of the learner.

7. Branching programme helps in the development of creativity and problem-solving ability.
8. Branching is most useful in the areas beyond facts, definitions and basic skills.
9. The frames being of larger size contain a good deal of information and this may enable the programmer to enrich his style and expand his ideas.
10. The programmed material in branching programme may be employed in the form of programmed text or teaching machines. The material when presented in a book form takes the form of a 'scrambled book'.

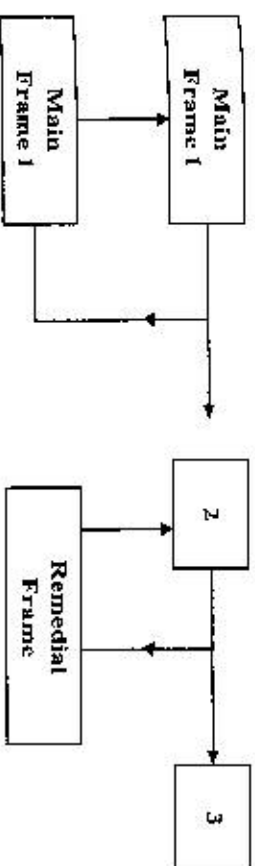
LIMITATIONS AND DRAWBACKS OF BRANCHING PROGRAMMING

1. The multiple choice questions provided in this programming may lead to guesswork on the part of the learner and he may not understand the subject matter of the frame.
2. The setting of appropriate multiple choice questions suiting to the entire material of the frames proves a difficult task.
3. No branching method can provide infinite branching to take care of all possible needs of every individual student.
4. The cost of branching programme is very high when compared with traditional teaching approaches.
5. The branching programme is not suitable for small children as they are unable to express the symbolism.
6. The programme needs revision after every two years.
7. It is difficult to cover the entire subject-matter.
8. The diagnostic questions framed by the programmer may or may not suit the needs of the individual learner.
9. The programme cannot shape the behaviour of the learner.

Backward Branching.

Backward branching is based on the principle of 'repeating the missed frames'. As shown in the diagram the learner of frame No. 1 of the main stream goes to frame No. 2 of the main stream only if he makes a correct choice. But if he makes a wrong choice, he is led to a remedial frame wherein he is given some more help in understanding the concept and in solving the solution by a

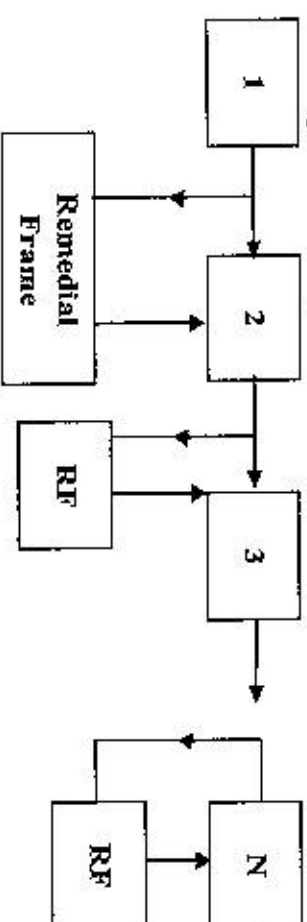
better logic. He will be then directed to the original frame No. 1 so that he can read it again and answer it correctly in the light of the remedial materials he has received. So the learner who has committed error goes through the same frame twice (once before the remedial material and once after the remedial material).



Forward Branching.

In the forward branching, whether the learner makes a wrong choice or correct choice, he will always be going to new pages, thus physically progressing from page to page.

The learner making a correct choice will go directly to the next frame of the main stream. But the learner who had made a wrong choice goes to a remedial frame wherein his mistake is fully explained, probably followed by another parallel question, from which he goes to the next frame of the main stream. So, even though he is wrong, physically he does not return to the original frame but progresses to a remedial frame which is more logical and simpler and then joins the main stream.



4.7.3 Programming and Models Comparison

Item/Concept	Branching	Linear	Mathematics
Proponent	Norman A. Crowder (1954)	R. F. Skinner (1954)	Thomas B. Gilbert (1962)
Learning theory	configuration theories of learning	Operant conditioning	Connectionist theory
Application	Remedial to the difficulties of the learner	Modification of behaviour	Mastery of the subject-matter
Stage utility	Very effective at higher classes	Very effective at the secondary stage	Useful for higher classes
Reinforcement	Ultimate correctness of response	Correctness of response	Task completion
Unit of presentation	The frame	The frame	The operant
Size of the step	Larger	Small	As big as possible
Number of steps	Small	Large	As few as possible
Response	Multiple-choice	Constructed	Written or by performing task on stimulator
Rate of error	Varies according to programme usually 20%	5%	Some deliberate errors written into the programme to sharpen discrimination
Subject-matter suitability	Where broader concept or large piece of information can be given	Subject which can be divided into small pieces	Suited to Kinesthetic or discrimination skills
Paper Programme	Scrambled text	Linear book	Programmed instructional package
Machine design	Film programme	Simple liner	Stimulation devices
Paper cost	Relatively expensive	Lower cost	High cost
Machine cost	More than linear	Cheap	Varies according to the type of stimulator

4.8 COMPUTER ASSISTED INSTRUCTION (CAI)

Computer assisted instruction (CAI), also known as Computer Based Instruction or Computer Managed Instruction, is the instructional method that uses a computer for transmission. CAI is used to enhance the quality of instruction and facilitate it. There are various CAI programs which incorporate tutorials, drill and practice, simulation, and problem solving approaches to render any topic understandable.

Most commonly CAI is rendered through Web-Based Tutorials (WBT). Academic libraries have incorporated web based tutorials in their inventories and have emerged as the popular way of instruction. WBT is an innovation popularized by the internet. The benefit of the WBT lies in the fact that, with it, the students learn any topic in a self-directed and self-paced way.

The efficacy of the WBT lie in how interactive they are. An instruction based on compendium of information cannot and should not pass for WBT simply by the virtue of being on the internet. Engagement of the student is the key and it should be interactive and stimulating enough to make up for the absence of the mentor. They can be a set of quizzes and have the provisions of feedback.

Information Literacy is another aspect that is emphasized by the promoters of WBT. The modules contained in the WBTs have an extensive range of topics that walk a student through a given topic while simultaneously generating information literacy.

CAI can also be viewed as a self-learning technique. It can be online or offline and involves material that promotes interaction of the student with programmed instructional materials. Through the computer the learning can be monitored.

CAI generally consists of a varieties of combinations of text, graphics, audio and video material to add value to the entire teaching-learning experience. It is being discovered globally, and gradually in Indian classrooms, that the computer can be used in various ways to assist students in accomplishing curricular goals.

What renders CAI so effective is the various ways of interacting and providing feedback. This is achieved with the use of text or multimedia content, multiple-choice questions, problems, immediate feedback, notes on incorrect responses, exercises for practice, worksheets and tests.

4.8.1 CAI-Meaning

Computer has contributed a lot in each and every sector of life. Computer assisted instruction (CAI) has emerged as an effective and efficient media of instruction in the advanced countries of the world. In fact, CAI is being used in formal and non-formal education at all the levels. In India too, computer has been introduced in most of the areas such as data processing, decision making. It has also impact on the working methods of research and development in the fields of science and technology. The computers are being used in the almost all areas of life such as transportation, communication, national defense, scientific research and education.

4.8.2 Focus of CAI

The main objective of programmed instructions is to provide individualized instruction just to fulfil the special needs of the individual pupil. In order to achieve this objective, some efficient device is required. This device should be flexible and it can store huge amounts of organized information. The device may help a person to use some selected part of the stored information. A computer fulfils all these requirements. It can store the organized information; it can process the information suiting to the needs of the individual learner. In short, CAI covers the entire educational system by providing itself a useful tool in teaching various subjects.

4.8.3 Principle of CAI

This strategy is based on the principle of machine on one hand and principles of programmed instruction on the other.

4.8.4 Purpose of CAI

The main purpose of CAI programme is to provide individual instruction to students according to their abilities and interests.

4.8.5 Structure of CAI

Both linear and branching types of instructional programmes are used in CAI. Students are directly connected with the computer via instructional material. The responses of the students are also recorded by the computer. On the basis of the responses, the computer takes decision as to which information is to be made available to the learner next. If learner's answer is correct, computer gives him feedback as well.

4.8.6 Steps involved in CAI

a. **Pre-tutorial phase:** The main purpose of the phase is that the student must achieve the objectives by using a specific type of

instruction. Here the objectives of instruction are first determined and then instructional programme is prepared and fed in the computer.

b. **Tutorial phase:** Its main focus is to select a proper instruction and present before the learner. It will control the learner's responses also. The learner may select one or more than one instructions at a time.

4.8.7 Origin of CAI

The first commercial computer was operative in 1951 in Census Bureau. First CAI attempt was made around 1961 when the University of Illinois is produced Programmed Logic for Automatic Teaching Operation (PLATO). Hence the use of computer in general education started from early sixties.

4.8.8 Various Instructional Modes

In the field of instructions a computer plays a major role. In these computer assisted instructions, it interacts directly with the learners while presenting the lessons. There are various instructional modes which can be facilitate by computer assisted instructions (CAI).

i. Tutorial Mode

In tutorial, information is presented in small units followed by a question the pupil's response is analyzed by the computer and appropriate feedback is given a network of branches or pathways can be programmed to teach. The pupils are allowed to work on their own pace.

ii. Drill or Practice

In this mode, the program leads the learner through a series of examples to develop dexterity and fluency using the skill. All correct responses are reinforced.

iii. Discovery Mode

Here, inductive approach is followed. The problems are presented and the pupils solve those problems through trial and error. It is just like laboratory learning. It aims at the deeper understanding of the results obtained from discovery.

iv. Gaming Mode

This mode may or may not be instructional, but it is recreational. Sometimes learning takes place through games. This mode is especially meant for young children.

v. Simulation Mode

Here the pupil faces scaled down approximation of real life situations. Hence, realistic practice takes place without involving any risk.

vi. Problem Solving Mode

Problem solving can be readily achieved provided the typical computational capability of the computer is available and there is a typewriter and display response device with remote control of two way communication. The students need to know how to communicate with computer and how to solve his problem.

vii. Inquiry Mode

Inquiry is mode of third type of CAI application. In this CAI system responds to student's inquiry with answers it has stored. In this mode instructional staff must learn how the system operates.

viii. Author Mode

CAI is used to support instruction by generating sets of materials for a student's use. In generating concept learning materials, these might be sentence forms which blanks in them each of which is to be filled by a word or a set of word, i.e., inserted into the blanks by computer according to the sets of instructions.

ix. Logo

This system was developed by Feurzeig and Papert at MIT. Logo is a simple programming language which can be taught to children. This programme provides instructions which can be used to produce pictures on screen. The children, who learn LOGO, make up their own programmes to draw flowers or faces or generate designs on the screen.

4.8.9 Role of teacher in CAI

CAI directly interacts with the students individually and with the teacher. Teachers are to play their role in CAI. Human teachers cannot be eliminated from teaching/learning process. We can highlight the role of a teacher in CAI in the following manner.

i. Use of New tools

CAI provides the teacher some chance to use new tools. This use will enhance the person's satisfaction. Also it will increase the individual's efficiency. It can produce elaborate graphs and drawings.

ii. Compatible with line teaching

CAI is compatible with line teaching. The teacher can devote his time for more creative activities.

4.8.10 Experts needed in CAI

Computer aided instructions need the help of the following experts:

- a. **Computer Engineer:** A computer engineer is a technical person and knows about basic principles and techniques of programme.
- b. **Lesson Writer:** The lesson writer is an expert who is familiar with lesson writing. Lesson writer may be experienced teachers or an experienced teacher may be a lesson writer.
- c. **System Operator:** He knows the system thoroughly and can cope with all commonly occurring failures of software and hardware of the system.

4.8.11 Uses of CAI

- ❖ It is the main sources of receiving facts and information's for the teachers and pupils.
- ❖ Drill and practice opportunities are provided to the pupils.
- ❖ It is useful to the form of learning laboratory.
- ❖ It is important in solving administrative problems.
- ❖ It is helpful in evaluation process.
- ❖ It is useful in framing time-table.

4.8.12 Limitations of CAI

- ❖ The computer fails to appreciate the emotions of students. The emotional climate created by teacher in direct class-room interactions with the students is absent in CAI.
- ❖ CAI programmes do not in themselves solve psychological or educational programmes.
- ❖ CAI fails to develop essential features of language competency where the ability to generate or construct meaningful sentences is essential.
- ❖ It was pointed out that some students got more tired than conventional study or felt like quitting the study.
- ❖ The peripheral equipment puts constraints in the ways on which a student can interact with the computer.

4.9 KELLER PLAN - PERSONALISED SYSTEM OF INSTRUCTION:

(PSI): The modern world is dynamic with its revolutionary changes in all spheres of human life. Needless to mention the great

changes have taken place rapidly with the introduction of educational technology the field of education and training in advanced countries of the world. The developing countries have also been imparting and adopting educational technology from the advanced countries who solve their own problem the class room communication has considerably been changed with application of education technology in teaching learning process with its emphasis on individualizing instructions. Personalized system of instruction (PSI) is the one of the recent innovation which has been successfully introduced in higher education to individualize instruction. This system of instruction which is person oriented. It is more emphasis on the individualization of instruction than other methods in higher education. The instruction is tailored to the need and ability of the individual learner. PSI get its name from the fact that each student is served as an individual by another person face to face and one to one in spite of fact that the class may contain number of students. It is suitable for courses for the student is expected to acquire a well-defined body of knowledge or skill. The majority of college course the PSI teacher expects almost all of his students to learn his materials well and is prepared to award high grades to those, who do, regardless of their relative in the standing in the class. The teacher accepts the responsibility meeting the goal within the normal limits of manpower, space and equipment.

4.9.1 Objectives of PSI:

The PSI has been to evolve to fulfill certain specific instructional objectives which may be enumerated as follows:

- ❖ To establish better personal – social relationship in the educational process.
- ❖ To provide frequent reinforcements for learning.
- ❖ To provide increased frequency and quantity of feed back to the instructors which the consequent benefit of a basis for meaningful revision in programme, content, and instructional procedures.
- ❖ To decrease reliance on the lecture for presentation or critical information utilizes different techniques for instructional purposes.
- ❖ To evaluate on the basis of fixed standard are mastery in a variable time period at the acceptable level of performances of the students.

4.9.2 Characteristics of PSI:

The personalized system of instruction (PSI) lays more emphasis on the importance of written works. The teacher gives practice to the learners on carefully prepared assignments consisting of section from standards text books. Articles are given to the students along with study question and other instructions as to what to read in what order and for what information. When the students things that he has mastered the materials he comes to the class room to take a brief quiz. This is immediately corrected by proctor. If there are errors the proctor indicates what part of the assignments needs further study the students goes off to do some more work and then come back to try again. That is are not examination in the normal sense. Students are not penalized for securing lower grade for an error. The specific distinctive characteristic of PSI is as follows.

- ❖ Self-pacing.
- ❖ Use of multimedia.
- ❖ Use of proctors.
- ❖ Mastery learning.
- ❖ Importance of written work.

4.9.3 Research on PSI:

Though PSI is very recent innovation in higher education, but it has stimulated a great amount of research in short life. A number of research studies have been conducted on the PSI model to study its effectively in instructional process in western countries. Very few research studies have been undertaken in our country to verify the clients of personalized system of instruction. The five important areas of instructional effectiveness which have been investigated by researchers are as follows:-

- ❖ Performance
- ❖ Retention
- ❖ Transfer of training.
- ❖ Efficient learning methods.
- ❖ Attitudes.

4.9.4 Current status of PSI:

The PSI has been tried out in all areas of education with great success. There are four major developments PSI, originally was developed to design a psychology programme and social science. All disciplines have tried PSI and there are PSI courses at all

levels of education from secondary through advanced graduate courses. The Second development is the implementation of PSI beyond the single course in to sequences of courses of entire college. The third development is the institutionalization of PSI. It has been successfully used in higher education and thousands of studies have been conducted in USA to establish in superiority over other current techniques are class room teaching.

4.9.5 Suggestions for implementing the PSI and its need in our educational system:

The PSI method is effective and yields better results than another comparable methods but the question is whether we may introduce it in our system of higher education. It is too early to predict its effective and efficiency in our country without conducting studies on the model of PSI. The method needs comprehensive planning and preparation before it is introduced. It should be experimented in some institute of higher learning and then gradually introduce in actual class room situation. After making modification, if needs be, the system should be introduced in higher education on large scale. The Second important factor is too prepared to teachers mentally to accept this innovation in higher education. The introduction of PSI entails extra academic work on the part of teacher which very few are ready to undertake in the present circumstances. The third problem is that the introduction of PSI necessitates the appointment of proctor who can establish face to face rapport with students which do not think if can afford at the present time. The undergraduate students may also we employed to access the quiz test and providing immediate guidance to the students. There is a great problem of providing TV, Tape Recorder, and Computer in addition to printed materials and discussions etc. The introduction of PSI needs restructuring of present classrooms which involves great financial liabilities which we are not prepared to take at present, since we are busy in solving other national problems on priority basis. The last but not least is the organization of orientation courses in which teachers may be trained to write course units on the PSI model.

4.10 PROJECT METHOD

The project method is an activity approach to learning as it provides activity for the learner as he tries to reach through his own obtained goals. In the project both the mental and physical activity leads to accomplishment after a task in a real situation in something concretes.

4.10.1 Definition

Projected has been defined as that form off co-ordinate activity that is directed towards the learning of the significant skill, or process Dr. Kilpatrick has defined it as a "whole hearted purposeful activity proceeding in a social environment". Stevenson says "it is a problematic act carried to completion in the natural setting".

According to Ballard "a project is a bit of real life that has been imparted into the school" paker define it as a "unit of activity in which the pupils are made responsible for planning and purposing".

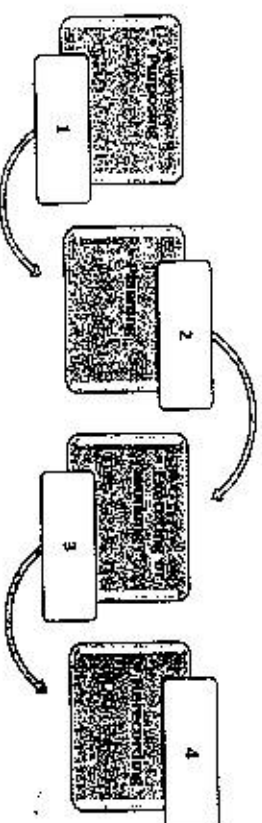
According to Richardson John S and Cahoon "there is nothing unique in term project its present use is simply to indicate a problem usually involving the use of the apparatus and materials, undertaken by an individual or a group. The problem may be simple and complex.

A project need not be new or original.

All these definitions imply that the term 'project' has got a very wide connotation and has been taken to include any activity like dramatic, pageants, making models, drawing maps and charts, collecting pictures, preparing scrap book going on field trips or any other constructive and experimental undertaking which enable the children to significant skill or process. Collins classifies project as

- 1) Exploration
- 2) Construction
- 3) Communication
- 4) Play
- 5) Skill.

4.10.2 Steps involved in a Project



1) Purposing

This phase is setting up a problematic situation and understanding exact problem. Motivation should be stressed here. Pupils should be given freedom to select the unit (project). The teacher should give suggestion and help the pupils to select the projects. While choosing problem the teacher should feel that it is the real need of students. They should know its purpose.

2) Planning:

Once a project is accepted the pupils are activated to get it done. So the teacher must emphasize planning. The entire responsibility should be given to the students. The art of good planning comes only by constant intelligent planning. Self-criticism is indispensable to planning and projects. The teacher should guide them by raising efficient questions. This can be done by discussions also every student should be encouraged to take part in the discussion.

3) Execution or working out:

This is the most important and interesting part of the project, since pupils are interested in doing things of a tangible nature. But here we have to guard against certain dangers.

- a) Some pupils may get themselves lost in the project and beings to forget the original objective with which it was chosen. The teacher should take effort to see that every phase of the activity is directed towards the final realizations of the objectives that brought the project into existence and see that child's attention and interest are not diverted from the educational aim.
- b) Sometime pupils may meet with difficulties in the working out of a project, and the tendency of the team or the individual may be to leave it partially done. A project should never be abandoned partly finished and the teacher should suggest way and means to surmount the difficulties and should be led to understand the practical value of seeing a thing to completion at whatever cost.

Judgment includes suggestions for improvement in what has been done. The improvement may be done by addition, elimination completion and modification for the various phases of the project judgments develops powers of critical analysis and ability to offer constructive criticism exercising a wholesome effort on the monotone of the group.

4) Recording:

All the students should put down a complete work of all the activities concerned with a project. This record will include.

- a) The choice of the project b) it's planning c) discussions held d) duties assigned e) references and books consulted f) difficulties felt g) experiences gained h) guidance sought, and i) important points for future reference and guidance.

4.10.3 Characteristics of a good project:

- i) It should be useful and purposeful.
- ii) It should provide for maximum number of activities.
- iii) It should provide such activities the material for which is available for reading.
- iv) It should be economical, it should not be expensive.
- v) It should provide for best learning experiences.

4.10.4 Advantage in the project Method:

- ❖ It entails no passive reception of information but enlist the activity of the pupils. They learn by doing.
 - ❖ It is of absorbing interest to the pupils. They think of nothing except their project and are always talking about the project.
 - ❖ It makes them work for principles and they acquire them in the process of evolving them.
 - ❖ It follows the psychological principles.
 - ❖ There is intimate connection between knowledge and conduct.
 - ❖ It makes the pupils to reason out and it develops reflective thought.
 - ❖ There is very little waste. The knowledge is acquired to satisfy a felt need.
 - ❖ Through project the students learn the school arts much more rapidly.
 - ❖ Law of readiness, exercise and effect are fully utilized in it.
 - ❖ Knowledge is gained in actual setting. It grows and flourishes in real life.
 - ❖ It trains the students for democratic way of life.
 - ❖ It emphasizes dignity of labour.
- 4.10.5 Disadvantages of Project Method:**
- ❖ It disorganizes the school work taking every one out into the open.
 - ❖ The knowledge acquired comes too scrappy. There is lack of systematizations.

- ❖ Correlation was designed to remedy the defect of extreme isolation, but in extreme correlation the demands of the subject might be neglected. The project idea puts everything into a common pool.
- ❖ It would lead to the random teaching. Formerly we selected the knowledge and then searched for application. Now we search for the knowledge.
- ❖ It would leave gaps in the pupil's knowledge; for certain portions of the subject never come into the project at all.
- ❖ It would be impossible for high schools. (Dr. Adams says that we may carry out one project every year)
- ❖ Project method is very expensive. Ordinary schools cannot afford this method.
- ❖ This method expects too much from the teacher. Every teacher cannot be expected to be efficient in each subject.

4.10.6 Conclusion and Suggestions:

- ❖ Projects should be used only to supplement the classroom teaching.
- ❖ To fill up the gaps in the knowledge learnt, direct lessons should be given.
- ❖ Projects should be based on co-operative activities and students should work in groups.
- ❖ Projects of income should be used in schools.
- ❖ Formal teaching should be provided for those subjects which are neglected by the project. E.g.:- practice in spelling, pronunciation and drawing should be given directly.

4.11 ACTIVITY BASED LEARNING (ABL)

What does activity based Learning classroom look like? Well... Active! Here, the teacher is mostly that of a facilitator rather than of authority in knowledge. He engages the learners into tasks and makes abstract ideas into concrete ones. It is learning by doing rather than learning by listening it can be done in two ways Student centered instruction where students have freedom to choose a problem and formulate strategies to solve them and

- i) Teacher centered instruction where the teacher takes the lead role.

The ABL approach is unique effective effort to attract out-of-school children to schools. The teachers who are involved in implementing this method have developed activities for each

learning unit which facilitated readiness for learning, instruction, reinforcement and evaluation. ABL has transformed the classrooms into hubs of activities and meaningful learning. Activity-based Learning is the method of education followed in the Corporation schools of Chennai (Tamil Nadu), from 2003 when M.P. Vijay Kumar became the Commissioner of the Chennai Conation.

The ABL concept has been taken from the Rishi Valley practices. This has been introduced in the Corporation Schools of Chennai with slight modifications. Seeing the success of the scheme this has been introduced in the Panchayat Union Schools.

4.11.1 The Process of ABL approach

- ❖ Each part/unit is called a milestone.
- ❖ In each subject, the relevant milestones are clustered and linked as chain and this chain of milestones is called LADDER.
- ❖ To enable the children to organize in groups group cards are used.
- ❖ Evaluation is inbuilt in the system. Separate cards / activities are used for this purpose.
- ❖ Each child is provided with workbook/worksheet for further reinforcement activities.
- ❖ Children's progress are recorded through annual assessment chart.
- ❖ Each milestone (Each part/unit is called a milestone) has different type of activities such as introduction, reinforcement, practice, evaluation, remedial and enrichment activities represented by different logos.

4.11.2 The Teacher's Role in ABL

The teacher has a very important role in this system. She has to learn the entire ABL system and work effectively with it. She has to exercise a quiet authority, without becoming authoritarian. An egalitarian attitude may require some un-learning and re-learning for teachers, but when they see it as part of the new culture of education, they are quick to accept it and practice it. They are also able, in this system, to spend some time on children who are slow.

4.11.3 Activity Based Language learning in the classroom

The activity-based learning methods have advantages as it appeals to a wide range of different senses and also appeal to

individual's varied learning styles. Different activities can be identified and can be tried in the skills thereby increasing the students' interest in learning process. The Activity Based Learning plays a key role in developing the critical and creative mind of students. Activities to learn Tenses, Prepositions, Speech, and Voices etc. are discussed upon and an analysis of the feedback is also made. Activities help students to reinforce the grammar techniques in their mind and also later relate them into real life situations.

4.11.4 An overview of Activity Based Learning Methodology (ABL)

To create an interactive atmosphere a language teacher can use varied activities in the classroom and this will make teaching and learning of language easy and spontaneous. In ABL method the learners have the opportunity to construct their own conceptualization and find solutions to problems. When the learner is exposed to a problem solving situation, meaningful learning takes place and the learner is able to identify, justify and defend his ideas.

In ABL method the teacher acts as a facilitator of learning who creates an environment for learners to question, exchange personal views and critically evaluate others' views and ideas. Here the role of the teacher in activity based learning is to develop the strategy of education by:

- ❖ The planning and preparation to Identify outcomes
- ❖ Having noted the outcomes of learning using proper strategies such as the agenda: report or wish to evaluate the status
- ❖ Select the appropriate activities to stimulate the learners
- ❖ Mechanisms within the group
- ❖ Promote cooperation in carrying out activity

In ABL method the learners have an opportunity for interactive learning

- ❖ Participate in educational objectives
- ❖ Collaborate with others to develop good organizational skills.
- ❖ Demonstrate enthusiasm for seeking new knowledge

4.11.5 Designing Classroom Activities

To make learning interactive and interesting, short activities should be included in the learning of language. This gives more scope for listening to English conversations, developing one's own

vocabulary and skill of speaking. Teachers should have an opportunity to apply the checklist as a way of deciding whether some activities are suitable for their teaching purposes.

Checklist for selecting or creating language-learning activities

Learning purpose Relevance to language learning Learning all language skills included? (Receptive and productive?)	Activity A (name of the activity /material used) Date :	Activity B
Learners		
❖ Difficulty level		
❖ Language level		
❖ Enjoyment/interest		
Management		
❖ Organization		
❖ Amount of noise		
❖ Type of material needed		
❖ Length of activity		
❖ Amount of preparation needed		
❖ space needed		
❖ Pressure to use English		
suitability for age group		
Learning principles		
Does it develop ... a positive atmosphere? .. Language creativity. Opportunities to experiment and notice language? ... Imagination? ... Purposeful and varied language use? ... Thinking skills?		

4.11.6 A Few Activities for Communicative Grammar Classroom

Brainstorming

Brainstorming activities give students the confidence to believe in themselves and their ability to speak English independently. It can be done individually on paper or in a group, orally or on the black board. It is a fast method of gathering ideas

for later connection and reflection. During a brainstorming session, learners share all the ideas that come to mind regarding the concept or topic under discussion. We can use Venn diagram, Spider Diagram, etc., to capture the ideas and facts presented by the learners. Fluency and quantity are important components of brainstorming. During a brainstorming session all ideas should be taken without criticism. Learner's s should try to build on their own and others' ideas to i.e. listen also to what their peers say.

Word/ Sentence Building

In this activity the teacher divides the class into two groups and makes use of synonyms; the word is contributed by one group and the other has to give the synonyms and vice versa to get points. The teacher conducts the activity and keeps the score. The same game can be played using antonyms, prefixes, suffixes etc., Activity based teaching paves the way for effective learning. Teaching should be student-centered and students must be motivated to participate in class activities. A word like 'save' ends with the letter 'e' and the next person in the group should say a word starting with 'e.' a word may be picked out which lists the related words. For example when the 'hospital' is written on the board and learners are asked to enlist the pertinent words, the related ideas that might come up are: doctors, patient, consultation, prescription, and diagnosis etc., A complete sentence can be given to learners and instruct them to add sentences related to the particular sentence to form a story. For example, - The passengers were sleeping in the bus when it hit -----

The Jigsaw Method

The Jigsaw is an active learning exercise. It is a useful strategy for studying narrative material. It helps learners and teachers effectively to cover areas of study material in a comparatively less time. In a large classroom it is a useful tool. In this method, (1) a general topic and learners are divided into smaller groups, (2) each member of the group is assigned to read and become an expert on a different piece of the puzzle (e.g., one person is given a Team Building Issues puzzle piece/article, another the Team Composition & Roles puzzle piece/article, and so on); (3) then, after each person has become an expert on their piece of the puzzle, they teach the other group members about that puzzle piece; and, finally, (4) after each person has finished teaching, the puzzle has been reassembled and everyone in the group knows something important about every piece of the puzzle. At the end the teacher

may also ask groups to give a presentation to the whole class about what they have learnt together



Just a Minute

In this activity the teacher guides the learners to frame sentences with--- errors and this will give them confidence and enthusiasm. The teacher gives few words and asks the learners to talk something on that word. The learner's may make errors but by practice the mistakes could be rectified. The teacher should always remember to appreciate their courage and willingness to participate in the class activities. At the end of the activity, the teacher should recap what each learner has contributed and appreciate his ideas, presence of mind and creativity. The teacher can guide learners to select the appropriate words without making them conscious of pointing to anyone in particular. This activity will not only make the learners concentrate on minor details while speaking but help them to become effective extempore speakers. It also involves the listeners' with utmost concentration and attentiveness.

Twenty Questions Game

This activity helps the learners develop the thinking and questioning skills. In this activity the teacher divides the class into two groups and decides the genre of their discussion. From the first group anyone group member can represent the group and they want to think a thing in their mind and they should not reveal their thinking to another group. By asking questions another group should find out the answers. The questions should be limited within twenty. Thus the students can develop the thinking and questioning skills. Through this activity, learners come out of their passive mode of listening, copying or taking notes they discover things for themselves and are likely to become more motivated to learn and respond better. Working on problems or tasks leads to greater independence among learners in their work and can act as appropriate pre-activity to instruct or input, preparing students to engage with -- but at the same time the material adopted should be appropriate for the learner's level.

4.11.7 Educational implications of ABL Method

A) Implications for students:

This study will:

1. Provide alternative method of understanding the concepts and grammar of English subject.
2. Show high significance to understand the opinion of the students towards use of activity based teaching strategy in English subject.
3. Help in making the students an independent learner

B) Implication for teachers: -

This study will help the teacher:

1. To improve their professional competency.
2. To develop all domains- cognitive, effective and psychomotor.
3. To create interest among students for their respective subjects.
4. To motivate them to develop various kinds of Activity based teaching strategies.
5. To enhance their self-esteem, confidence and to give them great sense of satisfaction in their profession.

C) Implication for Administration: -

This study will help:

To build a positive image of the educational institution due to improved quality of teaching and teachers.

To fulfil the objectives of the organization due to more satisfied students and teacher.

4.11.8 Recommendations of ABL Method

By adopting Activity based method of teaching the following criteria are used to observe student improvement.

- ❖ Increased understanding of the topic.
 - ❖ Improved interpersonal skills.
 - ❖ Willingness to participate in group activities.
 - ❖ Ability to relate the topic to real life examples.
- Day by day activity-based method of teaching is gaining its momentum due to its divergent utility in secondary schools.
- ❖ The subject teacher should analyze the content of their subject, identify the difficult topics and develop suitable

activity-based teaching strategy for those difficult concepts.

- ❖ The teacher should check the effectiveness of developed activity-based teaching strategy and modify it as per student's need.

- ❖ Activity based teaching strategies should be then included in teacher training program, so that the subject teachers can use it while teaching.

- ❖ Teacher should be encouraged to apply these programmes in their schools to enhance basic skills in English. Students should participate in the activities which will improve their skills.

4.11.9 Benefits/Features of ABL Method

- ❖ Children learn on their own pace.
- ❖ There is clarity in each lesson. Each lesson are planned in a systematic way. The child knows what must be done next. Each unit of information or process is broken up in such a way that clarity of the lesson is ensured.
- ❖ Classroom environment is highly enthusiastic.
- ❖ Children are truly engaged in the act of learning.
- ❖ There is wide scope for creativity of children.
- ❖ Provision of more time for self-directed learning and teacher directed learning is reduced considerably.
- ❖ Group learning, mutual learning and self-learning are promoted.
- ❖ Teachers teaching time is judiciously distributed among children. Only needy children are addressed by teachers.
- ❖ Children's participation in every step is ensured in the process of learning.
- ❖ Evaluation is inbuilt in the system it is done without the child knowing it.
- ❖ Rote learning is discouraged and almost no scope for rote learning. Periodical absence of child from school is properly addressed.
- ❖ Classroom transaction is based on child's needs and interests.
- ❖ Freedom to child in learning as he chooses his activity.
- ❖ Multigrade and multilevel in learning is effectively addressed.
- ❖ No child can move to the next higher step of learning unless attains the previous one.
- ❖ Sense of achievement boosts child's confidence and morale.

- ❖ Attractive cards and activity create interest among children.
- ❖ Scope for child's development in creative and communicative skills.
- ❖ Children will have a feel of security as they sit in rounds in the groups.
- ❖ Children are allowed to move in the classroom as they choose their activity.
- ❖ Moreover the distance between the teacher and the child is largely reduced and the teacher acts as a facilitator rather than teacher.

4.11.10 Advantages of ABL Methods

It helps learner to focus more and increase's attention and create an optimistic environment to respond to the educational settings for this different activities. The language activities can be identified and learners exposed to LSRW skills thereby developing their confidence levels, independent learning, enhance creativity, flexibility, assertiveness, leadership, team work and demonstrate a practical integration of knowledge, skills and aptitude. The learners are guided towards the development trends and their needs this method helps to develop their talents and the direction of education is correct. Self-study method and self-learning gives real opportunities where the benefits of the teaching learning situations in their future lead to the strengthening of the independence.

- ❖ The most important feature of activity based instruction is learning by doing. So this method of instruction can fulfil the natural urge of a growing child on one hand also can help them learn their lesson.
 - ❖ The method also promotes better understanding of a lesson among students as they learn the lesson by practicing the task themselves.
 - ❖ It inspires the students to apply their creative ideas, knowledge and minds in solving problems as well as promoting competitive spirit among them.
 - ❖ It also helps learner psychologically as they can express their emotions through active participation in something useful.
 - ❖ The method also helps in developing their personalities, social traits and inter-personal management skills.
- #### 4.11.11 Disadvantages of ABL Methods
- ❖ The activity based instruction method requires long-term planning with minute details of the whole process because

- before engaging the learners, the teacher has to make sure that all students have sufficient knowledge and skills regarding the task they are going to perform. So this method cannot be used on a regular and daily basis as it involves a lengthy procedure.
- ❖ The objectives of the method can only be fulfilled if the planning of the lesson is flawless. If there is slightest flaw in the planning, this method would do more harm than good.
 - ❖ Learners have varied levels of merit and understanding. So less meritorious students might not prepare for a task as other which might lead to failure of objectives of the whole process.
 - ❖ Many renowned educationists also are of the opinion that the activity based method is more suitable for branches of experimental sciences and less useful for subjects of social sciences.

Activity-Based Learning through Digital Schools

The main objective of this research paper is to acquire an understanding of activity-based learning through digital schools. As the name implies in activity-based learning, learners render an active participation in learning activities. Through the encouragement of activity-based learning, the educators are generating awareness among learners that if they participate in activities, then they will be able to augment their learning in an efficacious manner. In this type of learning, students are encouraged to participate in activities. In this manner, they become experienced and are able to acquire an efficient understanding of the concepts. Activity-based learning is promoted through digital schools through the organization of tasks, activities and projects. In order to implement activity-based learning in an effectual manner, the educators need to make provision of appropriate understanding among students and help them in achievement of desired academic outcomes. The main aspects that have been taken into account in this research paper include, meaning and significance of activity-based learning, measures to analyze activity-based learning, and major characteristics of activity-based learning.

Activity-based learning is the process, in which learners actively participate within the learning process. It is based on the core premise that learning should involve the implementation of experiments and activities as well, rather than just listening to lectures. The main factors that are taken into account in activity-based learning are, reading, writing, discussion, practical

activities, solving problems, analysis, synthesis and evaluation. It is also defined as any strategy that involves students in doing variety of tasks and thinking in terms of the tasks that they need to perform. Activity-based teaching is an approach that puts emphasis upon the idea that students should be encouraged to participate in actions. This is in contrast to some traditional forms of teaching, in which lectures are provided. The educators perform the role of the facilitators, and provide guidance and assistance to the students throughout the learning process (Hansraj, 2017).

Different methods are made use of by the educators to motivate students towards learning. Motivating the students and encouraging their participation is often accomplished through the implementation of various activities and projects. The project work, activities and tasks that are put into operation through digital schools have the primary objective of ensuring that students augment their learning and understanding. In this learning, educators are the ones, who need to put in their best efforts (Hansraj, 2017). They need to take into consideration various aspects, particularly when they are formulating tasks and activities. These include age groups of students, subjects and concepts, learning abilities, school and classroom environmental conditions, teaching-learning methods, teaching materials, resources, infrastructure and skills and aptitude. The tasks and activities need to be in accordance to the needs and requirements of the students. The educators need to ensure that they benefit from participating in them. Furthermore, they encourage teamwork, so students are able to enhance their communication skills and acquire opportunities to understand the concepts better.

Meaning and Significance of Activity-Based Learning

Learning is acquiring new knowledge, information and skills. Through the acquisition of these skills and information, individuals are able to carry out tasks and activities that are required to sustain one's living conditions sufficiently. Furthermore, it is the process that brings together cognitive, emotional and environmental influences and experiences for acquiring, enhancing or making changes to one's knowledge, skills or perspectives regarding the outside world. For activity-based learning to take place in an operative manner, it is necessary that students acquire an efficient understanding and participate in the tasks and activities that are associated with the acquisition of

information. Another aspect that is considered important is, students process the information with the establishment of high order thinking skills. These include, comprehension, analysis, synthesis, application and metacognition. These facilitate the acquisition of information among students regarding life activities. They are able to establish connection with learning that has taken place in the past, build own knowledge and enhance one's knowledge and information.

Activity-based learning is referred to the work that involves direct experience by the student rather than the textbook study. This learning should allow the students to engage with and process information in such a manner that would enable them to acquire effective understanding and augment knowledge in terms of particular concepts. An outcome of successful learning is that when the students acquire the basic literacy skills, they would be able to apply the knowledge and information in a relevant situation. The idea of activity-based learning follows the constructivist educational theory and is a child-centered pedagogy.

Activity-based learning is defined as the method of instruction, where activities are of different types. These are suitable, are in accordance to the subjects and are integrated into the regular instructional materials and methods that encourage the effectual participation of students in the teaching-learning processes. The primary objective of this learning is to make the classroom more student-friendly and students are less likely to feel intimidated.

Activity-based learning is regarded as the crucial element of the new thrust of what is commonly known as learner-centered or learning-centered teaching. Most of the definitions of activity-based learning encourages the participation of students in implementation of tasks and activities. When they are implementing tasks and activities, then they are able to develop creative thinking as well. Through this learning, improvements began to come about in the overall quality of teaching and learning processes. The students are able to generate numerous opportunities to clarify, apply, question and consolidate new knowledge. In this learning, it is the job duty of the educators to develop opportunities for the students to apply, and put into practice, modern and innovative techniques and methods. In other words, they act as leaders and mentors and guide the students in the right direction. When the students adequately listen and

follow the rules and guidelines, then they are able to acquire an efficient understanding of academic concepts and achieve academic goals in a suitable manner.

Measures to Analyze Activity-Based Learning

When activity-based learning is put into practice, it is vital to utilize measures to conduct its analysis. The main purpose of conducting the analysis is to ensure that they prove to be beneficial to the individuals in acquiring an efficient understanding of the academic concepts and achieving academic goals. The measures have been stated as follows:

Explanation of Lessons

Explanation of lessons is regarded as one of the most valuable aspects of activity-based learning. When the lesson plans are implemented, it is vital to ensure that structure is well-created and they are carried out in a systematic manner. When the educators begin with the lesson plans within the classrooms, they follow a systematic way of proceeding step by step through them. In this manner, the students also acquire an efficient understanding in terms of the stages of lesson plans. When students are provided with sufficient understanding of the concepts through activities, they are able to adequately understand the concepts.

Another major benefit is, activity-based learning help them in overcoming all the problems and challenges that are associated with learning. Mathematics is one of the subjects, in which activity-based learning help the students to acquire a better understanding of the concepts. When the students are encouraged to participate in activities, then they are normally motivated to work in a team and develop interactions with others. As it is apparent that when educators are formulating activities within the classrooms, they encourage students to collaborate and work in integration with each other. In this manner, they are able to enhance their communication skills and acquire an efficient understanding of the lesson plans. Through interaction, they are able to generate awareness in terms of viewpoints and perspectives of others. Hence, in this manner, they are able to produce better academic outcomes. Interactions are also regarded as beneficial, as students are able to clarify their doubts and overcome all the problems and challenges.

Classroom Environment

For activity-based learning to take place in an effective manner, it is vital for the educators and students to create and

amiable and pleasant classroom environment. Within the classroom environment, it is vital to possess adequate infrastructural facilities and resources, so the teaching and learning processes can take place in a well-organized manner. Research has indicated that in number of schools, proper furniture is not available and students are made to sit on the floor. Hence, within the classroom, when there are not any infrastructure, furniture, resources and facilities, then certainly there would be occurrence of impediments within the course of implementation of teaching-learning methods and achieving academic goals. Therefore, it is vital to ensure that within the classroom environment, there should be proper furniture, teaching-learning materials, resources, infrastructure and other facilities that would help the students and teachers in achieving academic goals. Within the classroom environment, it is vital for the teachers and students to work in collaboration and integration with each other. They need to create mutual understanding and form effective communication processes with each other. As the way individuals communicate and work with each other are one of the essential aspects of classroom environment. The teachers need to possess an approachable attitude and carry out their job duties in an appropriate manner. On the other hand, students need to develop interest and enthusiasm and pay regular attention towards their studies. When the teachers are diligent and put into practice effective teaching-learning methods, then students become motivated towards their studies. Whereas, when students complete their assignments on time, perform well in tests and pay adequate attention towards their studies, then teachers too feel pleased and satisfied with them. Therefore, the provision of materials, infrastructure and facilities and relationships are the aspects that constitute the classroom environment.

Participation of Students in the Learning Process

The primary objective of the students is to enhance their learning and achieve educational goals. To achieve this objective, they are required to focus upon augmentation of their skills and abilities. Bringing about improvements in competencies would enable them to render an effective participation within the learning process. The participation of the students within the learning process requires number of factors. These include, development of knowledge and skills, possession of interest, and enthusiasm, inculcation of traits of diligence, resourcefulness and

consciousness and effective communication skills. In order to enhance their participation in the learning process, it is vital for the students to pay adequate attention in class, particularly when the educators are teaching. When they give any assignments or ask the students to carry out any task or activity, then they should take it into consideration on an immediate basis. The students should avoid procrastinating work. When they are overwhelmed due to any problems or difficulties, then it is essential to interact with teachers or fellow students to provide their solutions. Therefore, participation of the students within the learning processes involves number of aspects. These include, paying attention in class, completing assignments on time, implementing effective time management skills, being regular in class, establishing mutual understanding and good terms and relationships with the teachers and fellow students, obtaining assistance from others to provide solutions to any difficulties or problems that they may experience, possessing adequate learning materials, such as, stationary items, text-books, notebooks etc. In higher educational institutions, students normally bring their lap-tops within the classroom setting. Apart from lap-tops, there is use of mobile phones and smartphones to a major extent in higher educational institutions. The students are making use of technology for numerous purposes for augmenting their learning and enriching overall quality of lives. Therefore, the implementation of these factors are regarded indispensable to encourage participation of the students within the learning processes.

Role of Educators

The role of educators is regarded to be of utmost significance in assisting and supporting the students in achieving their academic goals. The professional objective of the educators is to ensure that they implement teaching-learning methods in an appropriate manner, so they can promote effective growth and development of the students. The educators are the ones, who need to possess efficient skills and abilities to perform their job duties in a well-organized manner. Research has indicated that educators need to get involved in learning activities on a continuous basis. They participate in various training programs, workshops, seminars and conferences to generate awareness in terms of modern and innovative methods and approaches, which they can implement in their teaching methods. Furthermore, research is carried out on regular basis through the use of internet. Through

internet, they usually implement the tasks of research and writing and clarify their doubts in terms of any problems, which they experience, particularly regarding the academic concepts. Apart from imparting knowledge and information in terms of the educational concepts, the educators also act as mentors and guides. In some cases, they even interact with their students regarding personal problems and provide effective solutions. In educational institutions at all levels, the educators also interact with the parents. As parents need to remain updated in terms of the areas that their children are able to acquire an efficient understanding of the academic concepts, their growth and development takes place in an appropriate manner and they are able to achieve professional and personal goals. In some cases, educators and parents exchange ideas and suggestions, which would be beneficial in helping the students to enhance their learning and achieve academic goals. In order to perform their job duties in an appropriate manner, it is vital for educators to be professional and possess effective communication skills.

Implementation of Creativity and Resourcefulness

Within the system of education, the primary objective is to ensure that programs, methods, strategies and approaches should be formulated in such a manner that all the members are able to benefit. The educators should be able to carry out their job duties in an appropriate manner and incur job satisfaction. The administrative staff members also are able to render an efficient contribution in achievement of academic goals and feel satisfied with their jobs as well as the working environmental conditions. The rate of enrolment among students should increase, they should be motivated towards studies and generate good academic outcomes. These aspects can be observed, when there will be implementation of creativity and resourcefulness. The primary objectives of implementing creativity and resourcefulness is to ensure that all tasks and activities should be genuine, imaginative and generate productive outcomes. With the implementation of these aspects, individuals are able to feel, pleasurable and contented within the working environment. The various tasks and activities in which creativity and resourcefulness are put into operation include, teaching-learning methods, instructional strategies, organization of competitions, events, workshops and seminars, celebration of festivals and so forth. The individuals, primarily educators and staff members, who are involved in

organization of seminars, workshops and events, need to possess adequate skills, and abilities. The educators are responsible for the implementation of teaching-learning methods in accordance to the needs and requirements of the students, and subjects. Therefore, to implement creativity and resourcefulness, in some cases, even role play methods are put into practice. In primarily Hindi and English lesson plans, the students are asked to assume the roles of characters, they speak the dialogues and in this manner are able to acquire better understanding of the concepts. Hence, implementation of creativity and resourcefulness is regarded as beneficial in promoting enrichment of the overall system of education.

Major Characteristics of Activity-Based Learning

The major characteristics of activity-based learning have been stated as follows:

Cultural Values – The implementation of positive cultural values can lead to promotion of valuable learning situations that can lead to successful learning. In order to make effectual use of the activities to enhance teaching and learning methods, it is vital to take into consideration number of aspects. These include, formation of small learning groups, suitable teaching aids, appropriate methodology and well trained, qualified, skilled and practiced teachers (Superfine, n.d.). After one has acquired efficient information in terms of these aspects, they need to inculcate various activities. When there will be implementation of activities within teaching-learning methods and instructional strategies, then favorable environmental conditions would be created within classroom as well as the overall educational institutions.

Development of Curiosity – When the educators convey to their students that they are going to implement activities within the teaching-learning processes, in various subjects, such as, mathematics, science, social science, English or Hindi, then students normally develop curiosity among themselves. They feel curious that the educators are going to implement novice strategies and approaches to provide them adequate understanding of the lesson plans. In such cases, they ensure that they are not absent. As when scientific and innovative methods are put into operation, then students attend their class lectures on a regular basis. In activity-based learning, educators implement modern and innovative methods in activities that are related to lesson plans.

Intellectual Improvement – From the stage of early childhood, when parents get their children enrolled in pre-schools, they have the main objective of ensuring that apart from learning academic concepts, they should also be able to augment their intellectual skills. These include, wise decision making, rational thinking, problem solving and so forth. The children are taught that they should not be dependent upon their parents for each and everything and learn to make good and appropriate decisions themselves as well. For promoting intellectual improvement, the activities are put into operation by the educators. The students are encouraged to participate in these activities as a part of classroom instruction and augment their intellectual skills. Augmentation of Creativity – There are number of aspects that need to be taken into consideration, when understanding augmentation of creativity. The augmentation of creativity in teaching-learning methods, instructional strategies and in other tasks and activities of the educational institutions, normally is dependent upon the levels of educational institutions. In nursery schools, the school as well as the classroom environmental conditions turn out to be creative through the display of pictures, artworks and handicrafts that are prepared by the students. Whereas in secondary and senior secondary schools, creativity is promoted through putting charts and pictures within the classroom in terms of lesson plans and concepts. Whereas, in higher educational institutions, normally information is displayed, which may be useful to the students. It may be regarding programs, activities, seminars, workshops and so forth. Normally, students and professors collaborate and work in integration to promote the methods, which may augment creativity.

Development of Knowledge and Skills – Through participation in activities, the individuals are able to generate information not only in terms of academic concepts, but they are able to develop knowledge and skills in terms of other aspects. For instance, when students participate in role play activities to acquire understanding of English lesson plans, then they are able to develop their communication skills, build in self-confidence and enhance overall personality. On the other hand, when students are participating in music and dance activities, then too they usually tend to get involved in practice. Hence, they are able to enhance their creative skills. Development of communication skills,

creative skills and intellectual skills prove to be advantageous and meaningful to the students to a major extent.

Collaboration and Integration -- In activity-based learning, the educators and students are required to work in collaboration and integration with each other. When the educators make decisions in terms of the activities, then it is vital for them to communicate its benefits and all the information. The students need to be made informative in terms of how activities would prove to be meaningful and advantageous to them. The students are also given rights to give ideas and suggestions to the educators particularly, when they do not feel satisfied with any aspect of the activity. Hence, to generate desired outcomes through activity-based learning, it is essential for educators and students to work in collaboration and integration with each other.

Student-Centered and Learning-Centered -- There are numerous kinds of activities that are put into operation in accordance to the subjects, learning abilities, needs and requirements of the students, availability of materials, infrastructure and resources, skills and abilities of the educators and overall school and classroom environmental conditions. The examples of various activities are, sequencing, sorting, measuring, predicting, practicing, testing, memorizing, calculating, observing, role playing, dancing, singing, music, mixing, joining, tracing, painting, coloring, playing games, giving instructions and so forth (Superfine, n.d.). All these activities are student-centered as well as learning-centered. The educators instruct the students to put into operation these activities, primarily based upon the subjects that they are studying. For instance, in mathematics, activities, such as, sequencing, sorting, measuring, practicing and calculating are implemented. Whereas, in artworks, painting and coloring are implemented and so forth.

Task-Based Learning -- Learning usually tends to get monotonous, particularly when students are simply reading and memorizing the concepts. Hence, within the course of learning, the educators as well as the students ensure that task-based learning should be implemented on a frequent basis. In this type of learning, the students normally get involved in the implementation of tasks that would help them to acquire a better understanding of the academic concepts. The educators are the ones, who provide sufficient knowledge and training to the students in terms of

implementation of tasks. Whereas, it is the job of the students to ensure that tasks are carried out in an appropriate manner.

Generation of Pleasure and Contentment -- In some of the activities, such as role plays, dancing, singing, music, coloring and painting, students are required to work hard. But when they participate in these activities, they generate pleasure and contentment. When they are going to participate, they are required to augment their skills and possess confidence, which is necessary to enhance their performance. In higher educational institutions, mainly in colleges and university departments, when there are organization of workshops and seminars, then normally musical, singing and dance programs are organized. The students usually participate in them and receive certificates. In this manner, the participation of students in activities generates pleasure and contentment.

Improvement of Communication Skills -- Participation in activities require the students to get engaged in effective communication with other individuals. These primarily include their educators and fellow students. In educational institutions at all levels, learning cannot take place in seclusion. It is vital for the students to get engaged in effective communication with each other. In some cases, students possess an introvert nature. They simply attend their class lectures and do not tend to form effective communication skills with others. In such cases, educators normally encourage these students to participate in activities, where they are able to enhance their communication skills. When individuals need to generate awareness and augment their understanding in terms of certain aspects, then they are required to communicate with others.

Pedagogical Approach -- System of Delivery -- Governments have recognized that pedagogical methods can significantly enhance learning outcomes and with this perspective, they have progressively made use of variety of innovative pedagogic approaches. The activity-based learning approach is the approach that is adopted in schools in various parts of the country. This approach aims to make provision of involving and stimulating learning materials and a flexible space for learning through activity. The primary feature of this approach in learning is self-initiated, independent and takes place at an individual pace. Unlike the classroom environment in accordance to the required

standards, it allows for multi-age and multi-grade learning to take place (Aslam, Rawal, Vignoles, Duraisamy, & Shanmugam, n.d.).

Classroom Environment – The activity-based learning approach put emphasis upon learning without the burden of homework and without assessment through intimidating tests and examinations, as are put into practice in traditional classrooms. The physical structure of the classrooms in activity-based learning is different as compared to traditional classrooms in these classrooms, black-boards are available. The teachers normally make use of black-boards to provide explanation in terms of lesson plans and other academic concepts. In some cases, when proper furniture is not available, students are made to sit on the floor (Aslam, Rawal, Vignoles, Duraisamy, & Shanmugam, n.d.). The students normally make use of notebooks and write down notes, when the teacher is teaching. Furthermore, they are able to enhance their understanding through the use of textbooks. In some cases, information that is available within the textbooks is not adequately understood. In such cases, teachers encourage the students to make use of internet to augment their understanding.

Activity-based learning is a student-centered learning that is normally put into practice in schools in various parts of the country. In schools, this kind of learning approach is more commonly implemented as compared to higher educational institutions. As the term implies, this type of learning is primarily based on encouraging students towards participating in various activities. The educators utilize various types of activities within the teaching-learning methods and instructional strategies, so students are able to acquire an efficient understanding of the concepts and lesson plans and pursue towards achievement of their academic goals. The measures to conduct an analysis of activity-based learning are, explanation of lessons, classroom environment, participation of students in the learning process, role of educators and implementation of creativity and resourcefulness. When the educators are imparting knowledge and information in terms of lesson plans, then usually they need to make decisions in terms of activities that need to be put into operation to provide adequate explanation in terms of lesson plans and concepts. Furthermore, students should be encouraged to participate and there would be implementation of creativity and resourcefulness.

The major characteristics of activity-based learning are, cultural values, development of curiosity, intellectual

improvement, augmentation of creativity, development of knowledge and skills, collaboration and integration, student-centered and learning-centered, task-based learning, generation of pleasure and contentment, improvement of communication skills, pedagogical approach – system of delivery and classroom environment. In the last two aspects of pedagogical approach – system of delivery and classroom environment, the traits are highlighted in terms of how these differ in activity-based learning and traditional learning.

Research has indicated that students are able to benefit to a major extent from activity-based learning. In some cases, activities involve problem solving activities, whereas in others, the implementation of activities generates pleasure among students and motivate them towards learning. These normally include creative activities such as, singing, music, dance, artworks, handicrafts, role plays and so forth. Finally, it can be stated that through the implementation of activity-based learning, students develop motivation, interest and enthusiasm and render an effective participation towards achieving academic goals.

4.12 ACTIVE LEARNING METHOD (ALM)

Active learning is an approach to instruction that involves actively engaging students with the course material through discussions, problem solving, case studies, role plays, group projects, think-pair-share, peer teaching, debates, Just-in-Time Teaching, and short demonstrations followed by class discussion and other methods. Active learning approaches place a greater degree of responsibility on the learner than passive approaches such as lectures, but instructor guidance is still crucial in the active learning classroom. Active learning activities may range in length from a couple of minutes to whole class sessions or may take place over multiple class sessions.

4.12.1 Characteristics of ALM Method (Approach)

This method has been characterized by many things such as the development of skills: listening, speaking, reading and writing. Also, in this method the vocabulary is presented by dialogs that are learned through imitation and repetition, and grammatical rules are used by examples but it is no explicitly. Moreover, the target language is presented orally then teachers use the written presentation. When reading and writing are introduced learners have to report orally what they have read or write, and the

complicity of the readings and writings depends on the level of the students.

4.12.2 The 5 types of active learning

Examples of active learning activities include brainstorming, discussing, teaching, journaling, group work, focused listening, formulating questions, note taking, annotating, and roleplaying.

4.12.3 The elements of ALM in classroom

Each skill (listening, speaking, reading, and writing) is treated and taught separately. The skills of writing and reading are not neglected, but the focus throughout remains on listening and speaking.

4.12.4 Techniques That Would Be Use from This Method

Some techniques that we can use are the followings.

Chain drill: In these cases you can ask your learners to ask and to answer questions with a classmate. Another can be the transformation of drill: in this case the teachers can ask to their students to change the certain kind of sentence onto another one.

SWOT

Strengths	- Students learn the language automatically without stopping to think. - Students develop the five skills. - Students learn vocabulary through dialogs.
Weaknesses	- There is not meaningful learning. - there is too much repetition. - Students can lose their attention easily.
Opportunities	- Grammar is introduced by examples. - Fits you are touch how to say the words orally and then writing and reading. - Students develop the pronunciation skill.
Threats	- There is not creativity. - The method does not pay attention to learner's feelings. - Students are untrained in grammar.

4.12.5 Comparison of Traditional Teaching and Active Learning

Traditional Teaching	Active Learning
Professor lecturing	The focus of activities is the students
Students are passive learners	Students are active learners
Students watch and listen	Students solve problems, answer questions, discuss, and explain
Students work individually	Students work in groups
Cooperation is discourage	Cooperation is encourage

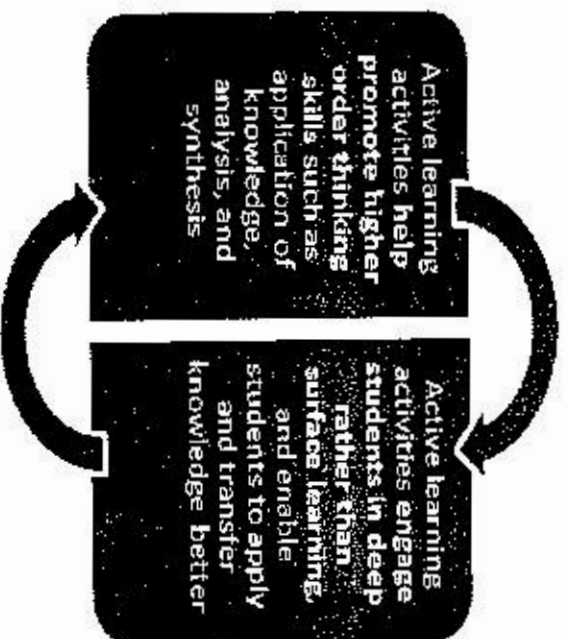
4.12.6 Teacher's and Learner's Role

In the teacher's roles the key is communication. The teacher models the target language. Since this method is teacher-dominated, the teacher is the one who controlled the direction and pace of learning and monitors and correct the learner's performance. Moreover, the teacher keeps the students attention using drills and tasks. The teacher chooses an important situation to practice the structure. Also, the teacher show to students how words relate to meaning in the target language, and get individual students to talk. In the case of student's role, since the teacher is a model for them. They have to imitate their teachers and follow the directions that the teacher gave them and have to answer as rapidly as possible. They are not allowed to use their native language since it is not good to master the foreign language.

4.13 ADVANCED ACTIVE LEARNING

4.13.1 Advanced Active Learning - meaning

Advanced active learning formats are designed to engage a student's critical thinking skills and apply previous and new knowledge to real-life scenarios. They still employ the critical elements associated with active learning "in doing things and thing about the things they are doing (reflection)" with more emphasis on the higher order learning processes.¹ They are processes designed for students to be responsible for their own learning with structured, and open-ended guidance from instructors.



4.13.2 Advanced Active Learning -importance

As you learned during the 5-minute Faculty Development Session on "Encouraging Classroom Participation", active learning involves student development of critical thinking and problem solving skills. More advanced engaged learning activities that require additional planning by the instructor are case-based learning, team-based learning or problem-based learning. Students continue to fail in their ability to demonstrate complex thinking skills because their educational experiences have provided little support for development of these skills for maximal performance. Multiple teaching styles exist to encourage students to apply critical thinking skills to open-ended problems utilizing multiple points of view.

4.13.3 Advanced Active Learning -Need

The traditional model of students as passive recipients of learning has been shown to be inadequate to foster a student's understanding and enhance a student's interest. As student's become active learners, faculty become active teachers. A teacher's responsibility moves beyond providing a didactic lecture and more towards encouraging students to be more independent after providing a conducive, team-based learning environment.

4.13.4 Advanced Active Learning Strategies

Case-based learning

Three-part case based learning, or scenario-based story, helps students develop a deeper understanding of the material. The instructor is responsible for creating a complex and engaging scenario or case which parallels the concepts of the lesson. Student may be additionally directed using guided questions however, students will use critical thinking skills to dissect the scenario while communicating a collaborating as a group. Using this technique, students are presented with real-life scenarios that are multi-faceted. Though these are often clinical scenarios, case based learning can be used in basic science courses to portray foundational knowledge that is applicable to clinical practice adding relevance to the material.

Team-based learning (TBL)

This approach utilizes student-student interaction in small teams to develop critical, practical, and creative teaching in their courses. Students assume the role of being inquirers and faculty burnout is decreased with increased student responsibility and student engagement. Key characteristics of TBL include the creation of heterogeneous work groups, a readiness process to ensure content knowledge, and small group activities. The team based learning module is especially appropriate in larger class sizes by allowing the class to function as multiple small classes.

Problem-based learning

In this approach, a students learning is facilitated by an instructor presented problem. Students will work through this carefully constructed, open-ended problem by engaging and applying new knowledge. The success of PBL is dependent on several factors: problems should allow for free inquiry, collaboration is encouraged and essential, and feedback in the form of reflection on their learning should be provided. The premise of PBL resides in the students to apply previous knowledge to new situations.

4.14 MIND MAPS

4.14.1 What is Mind Maps?

A mind map is a kind of realistic coordinator. Mind Maps permit you to make a visual representation of ideas on a specific

study point. They are especially helpful for brainstorming and note-taking.

They are an extraordinary note-taking and study instrument as they connect with students in entire brain thinking. They permit you to see the comprehensive view (how all the various ideas are associated with each other). They are utilized as a study help to help explain musings, decide connections, rearrange complex ideas, and remember information.

Have you ever studied a subject or brainstormed an idea, only to find yourself with pages of information, but no clear view of how it fits together?

This is where Mind Mapping can help.

Mind Mapping is a useful technique that supports learning, improves information recording, shows how different facts and ideas are related, and enhances creative problem solving. Read on to find out more.

4.14.2 What Are Mind Maps?

Mind Maps were popularized by author and consultant, Tony Buzan. They use a two-dimensional structure, instead of the list format conventionally used to take notes. This makes information easier to remember, as it's held in a format that our minds find easy to recall and quick to review.

A good Mind Map shows the "shape" of the subject, the relative importance of individual points, and the ways in which facts relate to one another. Research shows that this is of particular benefit when dealing with complex information, such as during business planning and strategy development.

Mind Maps are more compact than conventional notes, and often take up just one side of paper. This helps you to make associations and to generate new ideas. You can also add new information easily, even to a Mind Map that you've already drawn.

Mind Mapping can also help you to break down large projects or topics into manageable chunks, so that you can plan effectively without getting overwhelmed and without forgetting something important.

4.14.3 What Can I Use Mind Maps For?

Mind Maps are useful for:

- ❖ Brainstorming – individually, and as a group.

- ❖ Summarizing information.
- ❖ Taking notes.

- ❖ Consolidating information from different sources.

- ❖ Thinking through complex problems.

- ❖ Presenting information clearly.

- ❖ Studying and memorizing information.

Mind Maps are also good for refreshing information in your mind. When you commit the shape and structure of a Mind Map to memory, you can often get the cues you need to remember the information it contains just by glancing quickly at the Map. Studies have shown that this makes them highly valuable when you're learning a language, for example.

You can really get inventive with Mind Maps, so they are great for boosting creativity, too. When you include colors, images or drawings, they can even resemble a work of art!

4.14.4 Making mind maps for note-taking

Purposes behind Note-Taking

There are five significant purposes for taking notes:

1. Focus
2. Understanding
3. Recalling
4. Referral
5. Modification

4.14.5 Problem with Taking Notes Traditionally/ in Linear Way

All through our academic pursuits, from first grade through to graduation, we are determined what to gain and pursue knowledge however, the standard educational program hardly ever remembers tasks for how to learn. This is the reason such a large number of students despite everything invest their study energy featuring sections in textbooks and summing up parts when research has clearly indicated that those study techniques are the least compelling.

Essentially, linear notes are as yet the default group for note-taking for a great many people, even though it's from numerous points of view one of the least commonsense formats we might

pick. Here are only a couple of the most concerning issues with linear or traditionally notes taking way:

Linear notes can be incredibly monotonous, which makes them harder to memorize.

Linear notes don't give a decent review of a point. We frequently burn through significant time re-perusing huge lumps of our notes when searching for singular snippets of information.

When a page has been filled, it's very hard to add extra information to it. We are compelled to press it in the middle of the lines or include it at the base of the page with a reference bullet.

Furthermore, shouldn't something be said about taking notes in Word reports? Exploration has indicated "that while taking more notes can be useful, PC note takers' propensity to interpret addresses verbatim as opposed to processing information and rethinking it in their own words is unfavorable to learning."

Customary note-taking frequently feels relentless for us since it exhausts us intellectually while it debilitates us genuinely (take a stab at composing for two hours in a row and afterward perceive how your composing hand feels – most likely not extraordinary!)

4.14.6 How to Draw a Basic Mind Map

To draw a Mind Map, follow these five steps:

Step 1. Write the title of the subject or project that you're exploring in the center of a page and draw a circle around it, as shown in figure 1, below. (Our simple example shows a Mind Map of the actions needed to deliver a successful presentation.) (Figure 1.)



Step 2. Draw lines out from this circle as you think of subheadings of the topic or important facts or tasks that relate to your subject. Label these lines with your subheadings. (See figure 2, below.)

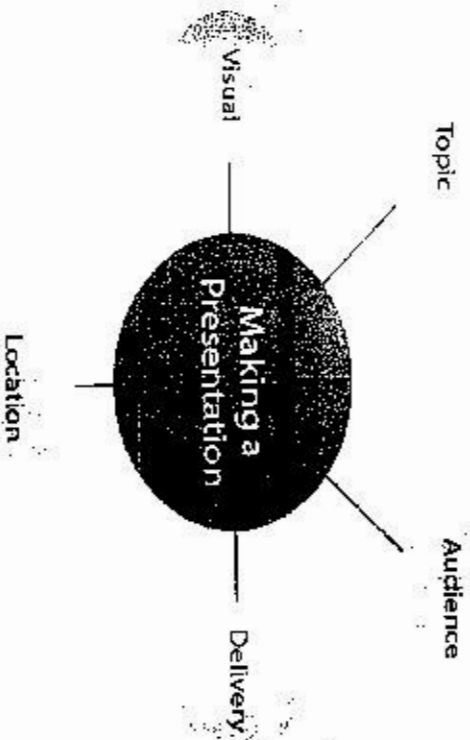
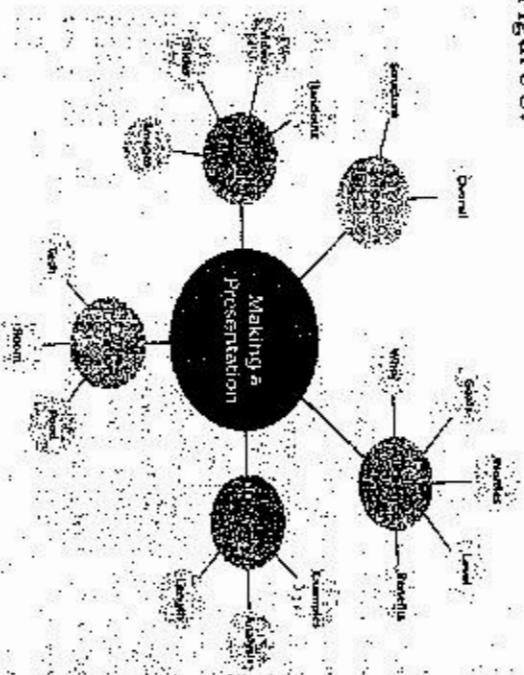
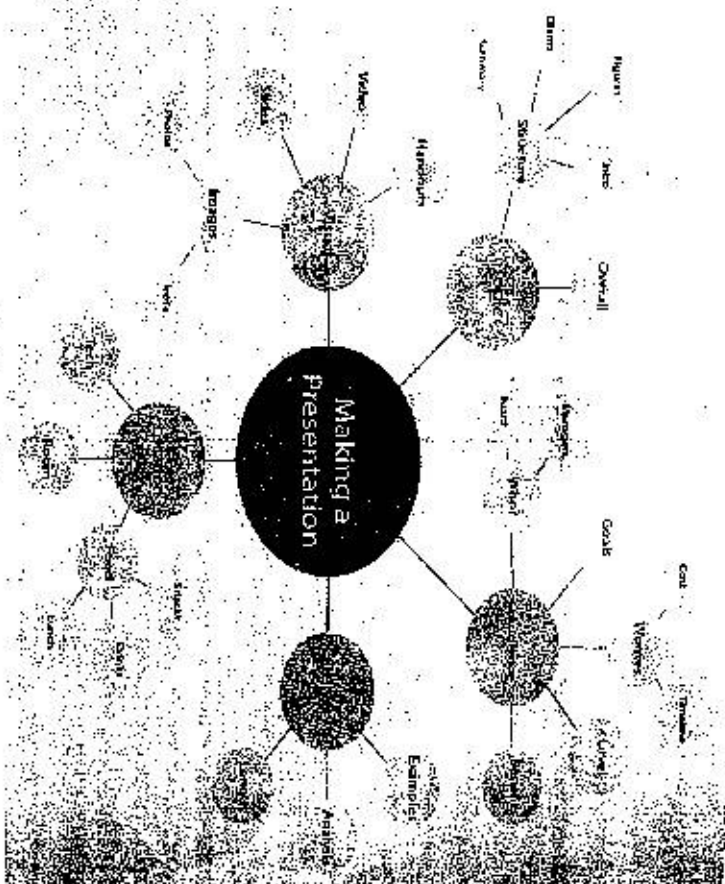


Figure 2.

Step 3. Dive deeper into the subject to uncover the next level of information (related sub-topics, tasks or facts, for example). Then, link these to the relevant subheadings. (See figure 3, below.) (Figure 3.)



Step 4. Repeat the process for the next level of facts, tasks and ideas. Draw lines out from the appropriate headings and label them, as shown in figure 4, below.



Bottom of Form

Step 5. As you discover new information or think of additional tasks, add them to your Mind Map in the appropriate places.

A complete Mind Map may have main topic lines radiating in all directions from the center, with sub-topics forking off these like branches and twigs from the trunk of a tree. You don't need to worry about the structure you produce - this will evolve of its own accord.

4.14.7 How to Use Mind Maps Effectively

Once you understand how to take notes in the Mind Map format, you can develop your own conventions for taking them further. The following suggestions can help you to get the most from your Mind Maps:

Use Single Words or Uncomplicated Phrases - Keep things simple. In Mind Maps, single strong words and short, meaningful phrases can convey the same meaning more potently. Excess words just clutter a Mind Map.

Print Words - They will be easier to read than joined-up or indistinct writing.

Use Color to Separate Different Ideas - Color can help to show the organization of the subject. It can also make your Mind Map a more appealing document, and help you to visualize the different sections of your Mind Map for future recall.

Use Symbols and Images - Pictures can help you to remember information more effectively than words, so use symbols or pictures that mean something to you, use it. (You can use photo libraries like Stock to source images inexpensively.)

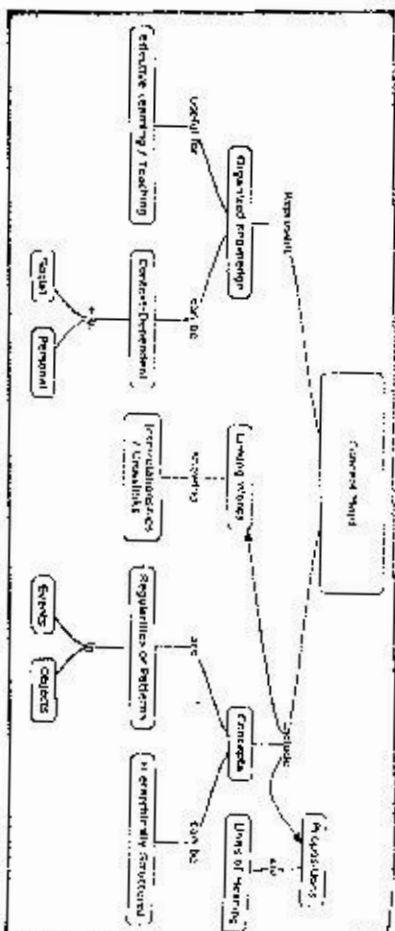
Using Cross-Linkages - Information in one part of a Mind Map may relate to another part, so draw lines to show these cross-linkages. This will help you to see how one part of the subject affects another.

4.14.8 Advantages of Mind Maps for Note-Taking

- ❖ Assists students with brainstorming and investigate any thought, idea, or issue
- ❖ Assists students with taking study notes and plan errands
- ❖ Encourages better comprehension of connections and associations among ideas
- ❖ Makes it simple to convey new ideas and manners of thinking
- ❖ Permits students to handily review information
- ❖ Makes it simple to arrange ideas and ideas

4.14.9 What is a Concept Map

A concept map is a visual presentation of information and details. Concept maps may have different designs such as tables, charts, graphic designs, flow charts, and Venn diagrams. The use of concept maps is an effective learning strategy since concept maps demonstrate facts visually. Thus, it will help students to memorize facts easily.



Moreover, concept maps provide students the opportunity to understand relationships more clearly. Concept maps can also be used to analyse information and compare and contrast them. Linked ideas and relationships are demonstrated clearly in concept maps. For example, we can demonstrate the process of making yogurt using concept maps, especially using a flowchart. This strategy of using a concept map not only works for school learners but also helps adults in workplaces to understand different topics and relationships. Concept maps help to encourage brainstorming and high-level thinking. At the same time, it gives clear communication of complex ideas.

Concept maps are primarily used for organizing and visualizing tacit knowledge, analyzing complex problems, identifying solutions, and taking action. They encapsulate a larger and more complex amount of information and are also used for explaining how these complex concepts relate to each other.

Therefore, it is important to emphasize that concept maps are more factual as they identify more main concepts and the systematic and complex relationships between them. One important difference between mind mapping and concept mapping is that concept maps include cross-connections between concepts that are described with 'action verbs' such as, contributes to, causes, requires, leads to, etc.

Concept maps are organized in a top-down hierarchical structure and are used to show multiple complex concepts and the relationships and cross-connections between them. In this regard, the topics featured in a concept map can have more parent and child topics while a mind map can have only one parent topic.

4.14.10 Difference between Concept Map And Mind Map

MIND MAP	CONCEPT MAP
A mind map is a diagram that presents data and information visually	A concept map is a diagram that represents the relationships between concepts and ideas
Has a radial structure	Has a tree structure with many branches and structures
Focuses on one concept	Focuses on multiple concepts
Creation is fast and spontaneous	Creation requires thinking skills
More flexible	More logical and maintain a hierarchical structure
Do not contain connection words	Contain connection words aligning with arrows
Easy to modify after creating	Not easy to modify since they have nodes

People often struggle to understand the real differences between concept mapping and mind mapping. Although they seem very similar, these two concepts differ from each other as they are represented by distinct structures and serve different purposes.

We will describe each mapping technique separately by emphasizing its unique characteristics, however, firstly, here is a little introduction about the origins of both mapping techniques.

Concept mapping is a technique that was introduced in the 1970s by the American professor and science researcher Joseph D. Novak. In particular, this technique was used for learning and teaching.

On the other hand, the first traces of diagrams similar to mind maps are noticed back in the 3rd century, however, the British psychology author Tony Buzan popularized the term 'mind mapping' in 1974. Mind mapping was used for 'mapping' information.

4.14.11 The Similarities of Concept and Mind Maps

Both mind mapping and concept mapping have individual narratives, which offer technical supports for their utilization. The two terms like this have precise meanings, anyway real or probable the cases behind them may be.

Mind maps are subjective, and clients are explicitly advised to "build up your very own style of Mind Mapping." **Concept maps** are the target presentation of the sensible connections among related ideas and thoughts or occasions.

Mind maps help create procedures as a strategy for brainstorming. **Concept maps** have an institutional capacity and fill in as a technique for introducing information and evaluating students' comprehension of information.

4.14.11.1 The Concept Map vs. Mind Map

Mind maps seem to establish concepts like advancements in thinking ready to enable universal programs to increase a severe edge. In this, they bring to mind the style and substance.

Concept maps are representations of knowledge, which show the concept and relationships among the multiple subjects. But unlike mind maps, concept maps usually use different shapes to display relations, instead of lines.

4.14.11.2 Differences in Definitions

Concept Map

- ❖ A system of interconnected concepts;
- ❖ Useful for information guide and substance hole examination;
- ❖ Mapping for implied information and substance that have many-to-numerous connections;
- ❖ More coherent;
- ❖ Unquestionable because of its association words;
- ❖ Difficult to change if more than 20 hubs.

Mind Map

- ❖ A tree graph for sorting out information;

- ❖ It is perfect for brainstorming thoughts and fast note-taking;
- ❖ It is for hubs that have straightforward connections, significantly coordinated;
- ❖ More adaptable;
- ❖ Non-questionable;
- ❖ Simple to alter in any event, for enormous maps.

4.14.11.3 Differences in Usages

Concept Map

- ❖ It assists youngsters with sorting out new information;
- ❖ It causes students to make essential associations between primary thought and other information;
- ❖ They're anything but difficult to develop and can be utilized inside any substance region.

Mind Map

- ❖ On Brainstorming – independently, and as a group;
- ❖ Summing up information;
- ❖ Taking notes;
- ❖ Uniting information from various sources;
- ❖ Thinking through complex issues;
- ❖ Introducing information unmistakably;
- ❖ Studying and retaining information.

4.14.11.4 Differences in Visual Structure

Concept maps look more doubtful and complicated, while mind maps are tree outlines generally. They have several spotlights on various components and elements.

Concept map esteems the two topics and associations. At the same time, mind maps look more on the subject themselves. That is the reason why **concept maps vs. mind maps** are essential to understand. If you look carefully, you believe that concept maps clarify all the more obviously on why individual hubs are associated. Then again, when you need to write down the same number of topics and information as you can, the mind map is a superior choice.

Concept map topics can be cross-connected. Be that as it may, mind map topics can't. So you can show many-to-numerous connections in concept mapping.
